
When Shift Happens - Confounding Is to Blame

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Abstract

Distribution shifts limit the robustness and external validity of machine learning models by altering the statistical information available for prediction. A dominant line of out-of-distribution (OOD) research treats causal-invariant representations as the principal route to robustness, yet recent empirical studies reveal a persistent and practically important pattern: (i) empirical risk minimization (ERM) can match or outperform specialized OOD generalization methods, and (ii) ERM often benefits from using all available covariates rather than only variables judged to be causal. We show that hidden confounding provides a coherent explanation for this behavior. Shifts in unobserved confounders induce coupled changes in observed covariate, label, conditional, and concept distributions, thereby weakening the observable invariances targeted by many OOD objectives. Under these conditions, our information-theoretic analysis shows that predictive information is maximized by preserving environment-specific relationships rather than by enforcing invariance alone. We further show that covariates acting as proxies for hidden confounders can improve generalization by changing the predictive-information terms that determine OOD performance. The resulting theory and experiments clarify why all-variable ERM can be strong in tabular OOD benchmarks and position covariate informativeness as a key consideration for OOD generalization under hidden confounding.

1 Introduction

Generalization—the capacity to draw reliable conclusions about unseen data from observed data—is a foundational objective across scientific and engineering disciplines. In medicine and the social sciences, it is formalized as *external validity*, the requirement that findings obtained in one population transfer to another [13]; in ecology, it motivates *space-for-time substitutions*, where spatial variation is used to reason about long-term temporal change [60]; and in engineering, it appears in *robust control*, where systems must retain performance under unmodeled disturbances [35]. Machine learning now provides a central computational framework for this objective, motivating a substantial literature on models that remain reliable beyond their training distribution [100, 88, 49].

For machine learning systems, generalization requires robustness to distributional discrepancies between training and test environments. A model trained to predict *food stamp reciprocity* from *household attributes* in one state, for example, should preserve predictive performance when deployed in another state with different policies and population composition. We use *out-of-distribution (OOD) generalization* to denote this requirement: a model should maintain performance on both in-distribution (ID) samples drawn from the training environments and OOD samples drawn from distinct test environments. Domain generalization [56, 31, 53, 99, 6, 38, 74], domain adaptation [98, 82, 51, 90, 91, 81], and robust learning [42, 69, 94] have consequently developed algorithms with guarantees under explicit assumptions on the form of shift.

A consistent empirical pattern challenges the sufficiency of this account. With careful model selection [30, 38, 87], standard *empirical risk minimization (ERM)* [86] often remains competitive with, and sometimes exceeds, specialized OOD methods on real-world applications [30, 24, 50, 57, 67, 87]. In addition, a benchmark study of 16 real-world tabular datasets reports that using all available covariates can improve OOD performance relative to restricting prediction to variables

judged to directly affect the outcome [57]. These observations are not merely anomalies in algorithm comparison; they indicate that existing theory does not fully characterize which predictive information should be preserved under realistic shifts.

Distribution shifts arise naturally when data are sampled from multiple environments [71]. In the food-stamp example, state-specific eligibility rules induce environment-dependent distributions of reciprocity and household characteristics. Much OOD theory responds to this heterogeneity by seeking *invariant relationships*: statistical associations between covariates and outcomes that remain stable across environments [6]. Learning such invariances is a central strategy for OOD generalization [6, 59, 31, 66, 56, 98, 63], and *causal invariance* provides an especially influential instance in which causal mechanisms remain unchanged across environments [71, 59, 6]. The difficulty is that real datasets rarely contain every variable relevant to prediction. Data-collection limits, privacy restrictions, and measurement error routinely leave important covariates unobserved [14, 52, 20, 48]. When omitted variables confound observed covariates and the outcome, shifts in their distributions can destroy the observable invariances on which these methods depend. Hidden confounding shift is therefore a structurally important failure mode: it can degrade OOD performance [40, 4, 85, 62] and can make a model learn an incorrect observed covariate–outcome relationship.

Despite its importance, hidden confounding shift remains only partially characterized in OOD generalization [4, 40, 85, 62]. This paper develops a unified theoretical and empirical account of how hidden confounding can explain the observed strength of ERM and all-variable prediction in recent benchmarks [30, 24, 50, 57, 67, 87]. Our contributions are as follows:

- We formalize hidden confounding shift as a source of coupled observable shifts and use this causal perspective to explain why informative non-causal covariates can improve prediction (§ 3).
- We derive an information-theoretic decomposition showing that, under hidden confounding shift, predictive information depends on environment-specific relationships; invariant representations remain useful but are insufficient by themselves (§ 4.1, § 4.2).
- We establish how variables informative about the outcome or the hidden confounders increase predictive information, thereby making principled covariate selection an explicit component of OOD generalization under confounding shift (§ 4.3).
- Through real-world and synthetic experiments, we show that (i) observable signatures compatible with hidden confounding appear across tabular OOD benchmarks, (ii) environment-specific predictive information is positively associated with ID and OOD accuracy, and (iii) informative non-causal covariates improve generalization in the studied settings (§ 5).

Overall, the paper clarifies why ERM can be unexpectedly strong under real-world distribution shifts and shows that hidden confounding is a structurally important source of both invariance-only failure modes and informative all-variable model success.

2 Related work

Out-of-distribution (OOD) generalization encompasses several related formulations, including domain generalization [56, 31, 53, 99, 6, 38], domain adaptation [98, 82, 51, 90, 91, 81], robust learning [42, 69, 94], federated learning [54, 36, 44, 97, 45], and OOD detection [41, 32, 92, 84, 49]. Domain adaptation assumes unlabeled test-domain data, whereas domain generalization typically uses environment labels only during training. Federated learning addresses multi-environment data through communication- and privacy-constrained collaboration, and OOD detection identifies samples that depart from the training distribution. Across these formulations, a recurring objective is to recover relationships that remain invariant across environments. Recent work shows, however, that invariance alone does not exhaust the requirements for generalization [46, 72, 93]. In particular, Ye et al. [93] emphasize that feature invariance must be complemented by feature *informativeness*. Our analysis makes this principle precise for hidden confounding shift by identifying when environment-specific informativeness, rather than invariance alone, determines predictive performance.

Proxy variable adjustment: When confounding variables are observed during training but unavailable at test time, Landeiro and Culotta [40] propose adjustment strategies that improve classifier performance under confounding shifts. Building on proxy-based adjustment for causal effect estimation [55, 39], recent domain-adaptation approaches recast unknown distribution shift as a causal-effect identification problem [4, 85]. OOD generalization under hidden confounding shift has also been analyzed under overlapping confounder support [62]. Our contribution is complementary: rather than

treating proxies only as an operational device, we characterize *why* proxy and informative variables improve OOD generalization by identifying the predictive-information terms they modify.

All variable models vs causal models. Recently, Nastl and Hardt [57] introduced a benchmark in which covariates are partitioned into causal, arguably causal, anti-causal, and other spurious groups. Across 16 datasets, models using all covariates Pareto-dominate models restricted to causal or arguably causal subsets on both ID and OOD performance. This empirical pattern is clear, but its theoretical explanation remains incomplete. We address that gap by identifying hidden confounding shift as a setting in which non-causal but informative covariates can legitimately increase predictive information. For linear causal models, anchor regression [68] provides a framework that interpolates between using all observed covariates and using only causal covariates. Our analysis instead explains the effect of adding covariates that need not be causal when hidden confounding is present. Eastwood et al. [21] show that unstable covariates can improve performance when they carry label information and are conditionally independent of stable covariates given the label, and they adjust shift using unlabeled test-domain data. On a medical dataset not specifically designed for that setting [8], however, ERM remains competitive, consistent with Nastl and Hardt [57]. This is also compatible with the broader result that, under well-specified covariate shift, maximum likelihood estimation (MLE) is minimax optimal for OOD generalization [26]. The settings we study are less well specified: hidden confounding shift couples multiple observed shifts, and the all-variable advantage arises from predictive information carried by observed proxies.

3 Manifestations of hidden confounding shift

This section formalizes how hidden confounding shift appears in observed data and why it creates a first-order obstacle for OOD generalization. We connect standard shift categories to a latent-variable causal account and then use a simple causal-effect identification example to show how observed covariate–outcome relationships can become misleading when the confounder is unobserved.

Types of distribution shifts. For covariates $\mathbf{X}$ and target Y , two environments e and e' can differ through any of the observed distributions summarized in Table 1. A common generative source of these differences is a change in the distribution $\mathbb{P}(U)$ of an unobserved variable U that causes $\mathbf{X}$, Y , or both (Figure 1). Such latent shifts propagate through the causal system and appear as observed distributional discrepancies over $\mathbf{X}$ and Y . For example, $\mathbb{P}(\mathbf{X}) = \sum_U \mathbb{P}(U)\mathbb{P}(\mathbf{X} | U)$, so $\mathbb{P}^e(U) \neq \mathbb{P}^{e'}(U)$ together with $U \rightarrow \mathbf{X}$ yields $\mathbb{P}^e(\mathbf{X}) \neq \mathbb{P}^{e'}(\mathbf{X})$. Analogous arguments apply to $\mathbb{P}(Y)$, $\mathbb{P}(\mathbf{X} | Y)$, $\mathbb{P}(Y | \mathbf{X})$, $\mathbb{P}(\mathbf{X}, Y)$, showing that a single hidden shift can generate several distinct observable shift categories.

Table 1: Distribution-shift categories expressed through environment-specific observed distributions.

Type of Shift	Mathematical Expression
Label	$\mathbb{P}^e(Y) \neq \mathbb{P}^{e'}(Y)$
Covariate	$\mathbb{P}^e(\mathbf{X}) \neq \mathbb{P}^{e'}(\mathbf{X})$
Conditional Covariate	$\mathbb{P}^e(\mathbf{X} Y) \neq \mathbb{P}^{e'}(\mathbf{X} Y)$
Concept	$\mathbb{P}^e(Y \mathbf{X}) \neq \mathbb{P}^{e'}(Y \mathbf{X})$
Joint	$\mathbb{P}^e(\mathbf{X}, Y) \neq \mathbb{P}^{e'}(\mathbf{X}, Y)$

Many OOD methods rely on causal structures among U , $\mathbf{X}$, Y that imply specific observable invariances. If $U \rightarrow Y \rightarrow \mathbf{X}$ (Figure 1 (a)), then $\mathbb{P}(\mathbf{X}, Y) = \mathbb{P}(Y)\mathbb{P}(\mathbf{X} | Y)$; an observed dataset shift $\mathbb{P}^e(\mathbf{X}, Y) \neq \mathbb{P}^{e'}(\mathbf{X}, Y)$ can arise from label shift, $\mathbb{P}^e(Y) \neq \mathbb{P}^{e'}(Y)$, while the conditional covariate distribution remains invariant, $\mathbb{P}^e(\mathbf{X} | Y) = \mathbb{P}^{e'}(\mathbf{X} | Y)$, because $\mathbb{P}(\mathbf{X} | Y, U) = \mathbb{P}(\mathbf{X} | Y)$. If $U \rightarrow \mathbf{X} \rightarrow Y$ (Figure 1 (b)), dataset shift instead appears as covariate shift, $\mathbb{P}^e(\mathbf{X}) \neq \mathbb{P}^{e'}(\mathbf{X})$, while $\mathbb{P}^e(Y | \mathbf{X}) = \mathbb{P}^{e'}(Y | \mathbf{X})$ remains stable [70, 16, 80, 11, 28]. These observable invariances need not survive the more general case in which U causes both $\mathbf{X}$ and Y (Figure 1 (c)) [50, 4, 40, 85, 62, 65, 64]. When $\mathbb{P}(U)$ shifts across environments, label shift, covariate shift, conditional covariate shift, and concept shift may occur simultaneously. The remaining invariance is conditional on the unobserved confounder, $\mathbb{P}^e(\mathbf{X}, Y | U) = \mathbb{P}^{e'}(\mathbf{X}, Y | U)$, a condition known as *stable confounding shift* [85, 4]. Because U is hidden, this invariance is not directly accessible from observed data.

Information-theoretic measures: Following Federici et al. [22], we quantify distribution shifts with mutual information. Let E denote the environment variable. We use $I(\mathbf{X}; E)$, $I(Y; E)$,

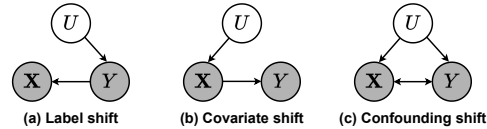


Figure 1: Causal graphs illustrating how latent variables induce commonly studied distribution shifts.

If $\mathbb{P}(\mathbf{X} | Y, U) = \mathbb{P}(\mathbf{X} | Y)$ [89, 83, 25, 5, 47, 96, 18, 79]. If $U \rightarrow \mathbf{X} \rightarrow Y$ (Figure 1 (b)), dataset shift instead appears as covariate shift, $\mathbb{P}^e(\mathbf{X}) \neq \mathbb{P}^{e'}(\mathbf{X})$, while $\mathbb{P}^e(Y | \mathbf{X}) = \mathbb{P}^{e'}(Y | \mathbf{X})$ remains stable [70, 16, 80, 11, 28]. These observable invariances need not survive the more general case in which U causes both $\mathbf{X}$ and Y (Figure 1 (c)) [50, 4, 40, 85, 62, 65, 64]. When $\mathbb{P}(U)$ shifts across environments, label shift, covariate shift, conditional covariate shift, and concept shift may occur simultaneously. The remaining invariance is conditional on the unobserved confounder, $\mathbb{P}^e(\mathbf{X}, Y | U) = \mathbb{P}^{e'}(\mathbf{X}, Y | U)$, a condition known as *stable confounding shift* [85, 4]. Because U is hidden, this invariance is not directly accessible from observed data.

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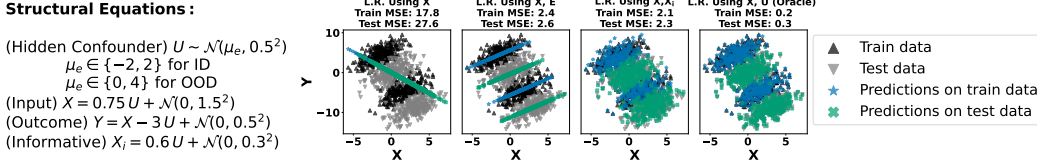


Figure 2: Four linear regression (L.R.) models evaluated under hidden confounding shift with minimal environment overlap (distant μ_e). (i) Training only on X learns the wrong X - Y relationship, illustrating Simpson’s paradox. (ii) Environment-specific summary statistics of X , denoted E , recover the correct relationship but have limited representational capacity. (iii) An informative covariate X_i for U improves OOD generalization. (iv) The oracle model is trained on X and U .

151 $I(\mathbf{X}; E|Y)$, $I(Y; E|\mathbf{X})$, $I(\mathbf{X}, Y; E)$ to measure environment dependence in $\mathbb{P}(\mathbf{X})$, $\mathbb{P}(Y)$, $\mathbb{P}(\mathbf{X} |$
152 $Y)$, $\mathbb{P}(Y | \mathbf{X})$, $\mathbb{P}(\mathbf{X}, Y)$, respectively. Many OOD objectives effectively impose one of two
153 simplifying conditions: $I(\mathbf{X}; E | Y) = 0$ under label shift or $I(Y; E | \mathbf{X}) = 0$ under covariate shift.
154 Hidden confounding shift removes this simplification because both quantities can be nonzero, so no
155 single observed invariance summarizes the shift.

156 **Challenges with hidden confounding shift:** Figure 2 illustrates the mechanism by which hidden
157 confounding shift obstructs OOD generalization. The outcome Y depends on the observed covariate
158 X and an unobserved confounder U , and environments are generated by shifts in the mean of U ,
159 denoted μ_e . A linear regression model trained only on X recovers the wrong association between X
160 and Y , as shown in the first subplot. This is Simpson’s paradox [73]: within each environment Y
161 increases with X , but the pooled relationship has the opposite direction. We observe the same qualita-
162 tive failure for OOD-oriented methods including IRM [6], VREX [38], and Group DRO [69]. In § 4,
163 we show theoretically that the appropriate strategy in this setting is to learn environment-specific
164 relationships, which can recover the correct X - Y relation. The second subplot demonstrates this
165 effect using environment-specific summary statistics of X , including means, standard deviations,
166 and quantiles. These statistics play a role analogous to *backdoor adjustment* in causal effect esti-
167 mation [58], because they proxy for information needed to adjust for the hidden confounder. The
168 same panel also shows that environment information recoverable from observed covariates may have
169 insufficient representational capacity. As § 4.3 establishes, additional informative covariates that
170 proxy for U further improve generalization, as shown in the third subplot. The oracle model, trained
171 on both X and U , provides the best-possible mean squared error for comparison.

172 4 Impact of hidden confounding shifts on OOD generalization

173 This section develops the information-theoretic account of how hidden confounding shift changes the
174 objective of OOD generalization. Let $\hat{Y} = (f \circ \phi)(\mathbf{X})$ denote the prediction for input $\mathbf{X}$, where f is
175 a classifier and ϕ is a feature extractor or representation map. We write $H(X) = -\mathbb{E}_X[\log(\mathbb{P}(X))]$
176 for entropy and $I(X; Y) = \mathbb{E}_{X,Y}[\log \frac{\mathbb{P}(X,Y)}{\mathbb{P}(X)\mathbb{P}(Y)}]$ for mutual information. The environment-specific
177 risk of $(f \circ \phi)$ is $\mathcal{R}^e(f \circ \phi) = \mathbb{E}_{(\mathbf{X}, Y) \sim \mathbb{P}_{\mathbf{X}, Y}^e}[\ell((f \circ \phi)(\mathbf{X}), Y)]$, where $\ell(\cdot, \cdot)$ is a loss function.
178 OOD generalization seeks predictors that perform well on both ID and OOD data, and the literature
179 operationalizes this goal through several objectives. Robust optimization methods [7, 9, 19, 75, 69]
180 minimize the worst training-environment risk, $\mathcal{R}^{\text{Rob}}(f \circ \phi) = \max_{e \in \mathcal{E}_{tr}} \mathcal{R}^e(f \circ \phi)$, where $\mathcal{E}_{tr}$ is
181 the set of training environments. Variance risk extrapolation (VREX) [38] minimizes $\mathcal{R}^{\text{VREX}}(f \circ$
182 $\phi) = \sum_{e \in \mathcal{E}_{tr}} \mathcal{R}^e(f \circ \phi) + \beta \times \text{Variance}(\{\mathcal{R}^1(f \circ \phi), \dots, \mathcal{R}^{|\mathcal{E}_{tr}|}(f \circ \phi)\})$, with hyperparameter β .
183 Invariant risk minimization (IRM) [6] minimizes $\sum_{e \in \mathcal{E}_{tr}} \mathcal{R}^e(f \circ \phi)$ subject to a classifier f that is
184 simultaneously optimal across environments. Empirical risk minimization (ERM) pools all training
185 environments and minimizes $\mathcal{R}^{\text{ERM}}(f \circ \phi) = \sum_{e \in \mathcal{E}_{tr}} \sum_{(\mathbf{X}, Y) \in \mathcal{D}^e} \ell((f \circ \phi)(\mathbf{X}), Y)$, where $\mathcal{D}^e$ is
186 the observed dataset from environment e [6, 38].

187 To compare these objectives under hidden confounding shift, we analyze the predictive information
188 $I(Y; \hat{Y})$ with $\hat{Y} = (f \circ \phi)(\mathbf{X})$. This quantity measures how much the prediction reduces uncertainty
189 about the true label and therefore provides a common information-theoretic target for prediction.
190 The analysis begins with two representation properties that are central to OOD generalization:
191 *informativeness* and *invariance* [93].

192 **Definition 4.1** (Informativeness and Conditional Informativeness). *The informativeness of fea-*
193 *tures $\phi(\mathbf{X})$ for predicting Y is defined as $\text{INF}(\phi(\mathbf{X}), Y) = I(\phi(\mathbf{X}); Y)$. The conditional informa-*

194 *tiveness of features $\phi(\mathbf{X})$ for predicting Y conditioned on environment variable E is defined as*
 195 *$\text{CINF}(\phi(\mathbf{X}), Y, E) = I(\phi(\mathbf{X}); Y|E)$.*

196 Conditional informativeness is the label-relevant information retained by $\phi(\mathbf{X})$ within each envi-
 197 ronment. Reducing CINF necessarily discards predictive information, although maximizing it can
 198 be inappropriate in settings such as algorithmic fairness, where the representation may exploit sensi-
 199 tive or biased within-environment signals. Such dependence can be discouraged by minimizing
 200 $I(\phi(\mathbf{X}); E)$, but doing so may reduce predictive performance, as shown in § 4.2. In the setting
 201 considered here, where E is not treated as sensitive information, maximizing CINF is a legitimate
 202 route to improved generalization, and the decomposition below makes this role explicit.

203 **Definition 4.2** (Variation and Invariance). *For a given label Y , the variation in the features $\phi(\mathbf{X})$*
 204 *across environments E is defined as $\text{VAR}(\phi(\mathbf{X}), Y, E) = I(\phi(\mathbf{X}); E|Y)$. For a given label Y , the*
 205 *invariance of the features $\phi(\mathbf{X})$ across environments E is defined as the negative of the variation i.e.,*
 206 *$\text{INV}(\phi(\mathbf{X}), Y, E) = -\text{VAR}(\phi(\mathbf{X}), Y, E) = -I(\phi(\mathbf{X}); E|Y)$*

207 Invariance measures the stability of $\phi(\mathbf{X})$ across environments after conditioning on the label. Low
 208 invariance permits environment-specific information to remain in the representation and can induce
 209 overfitting, whereas high invariance suppresses such dependencies and supports invariant learning.
 210 Extending the information-theoretic shift measures from § 3, we use $I(\phi(\mathbf{X}); E)$ for *feature shift*
 211 and $I(Y; E | \phi(\mathbf{X}))$ for *concept shift*. Depending on whether the original covariates or learned
 212 representations are under discussion, *concept shift* refers to $I(Y; E | \mathbf{X})$ or $I(Y; E | \phi(\mathbf{X}))$.

213 4.1 A general decomposition of predictive information

214 We first recall the causal-graph notation used in the decomposition. A causal graph $\mathcal{G}$ has nodes
 215 representing random variables and directed edges representing direct causal influence. A *path* between
 216 nodes X_i and X_j is a sequence of unique edge-connected nodes, and a *directed path* from X_i to X_j
 217 with $i < j$ has all arrows oriented toward X_j : $X_i \rightarrow X_{i+1} \rightarrow \dots \rightarrow X_{j-1} \rightarrow X_j$. Along such a
 218 path, X_i is a *parent* of X_{i+1} , X_{i+1} is a *child* of X_i , X_i is an *ancestor* of X_j , and X_j is a *descendant*
 219 of X_i . The three elementary path structures are chains $X_i \rightarrow X_j \rightarrow X_k$, forks $X_i \leftarrow X_j \rightarrow X_k$,
 220 and colliders $X_i \rightarrow X_j \leftarrow X_k$. In chains and forks, X_i and X_k are marginally dependent but
 221 become conditionally independent given X_j , i.e. $X_i \not\perp\!\!\!\perp X_k$ and $X_i \perp\!\!\!\perp X_k | X_j$. In colliders, X_i
 222 and X_k are marginally independent but become conditionally dependent after conditioning on X_j
 223 or any descendant of X_j , i.e. $X_i \perp\!\!\!\perp X_k$ and $X_i \not\perp\!\!\!\perp X_k | X_j$. A path from X_i to X_k is *blocked* by a
 224 conditioning set $\mathcal{S}$ if it contains a chain or fork whose middle node is in $\mathcal{S}$, or if it contains a collider
 225 such that neither the collider nor any descendant is in $\mathcal{S}$. If $\mathcal{S}$ blocks every path from X_i to X_k , then
 226 $X_i \perp\!\!\!\perp X_k | \mathcal{S}$; otherwise, the path remains open.

227 Now consider predictive information $I(Y; \hat{Y})$, where $\hat{Y} =$
 228 $(f \circ \phi)(\mathbf{X})$ is computed from the learned representation $\phi(\mathbf{X})$.
 229 Figure 3 represents the causal relations among $\mathbf{X}, Y, U, \phi(\mathbf{X})$,
 230 $\hat{Y}$, and E , with U denoting the hidden confounder. Conditional
 231 on either $\mathbf{X}$ or $\phi(\mathbf{X})$, the prediction $\hat{Y}$ adds no additional in-
 232 formation for reasoning about Y ; formally, $\hat{Y} \perp\!\!\!\perp Y | \mathbf{X}$ and
 233 $\hat{Y} \perp\!\!\!\perp Y | \phi(\mathbf{X})$. The environment variable E indexes shifts in
 234 $\mathbb{P}(U)$ across environments. *We first derive a general decompo-*
 235 *sition of predictive information without specifying the causal*
 236 *effect of U on $\mathbf{X}$ and Y . Section 4.2 then specializes the analysis to hidden confounding structures.*

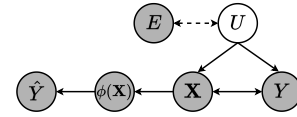


Figure 3: The bi-directed arrow between $\mathbf{X}$ and Y indicates that some covariates in $\mathbf{X}$ can cause Y , while others may be caused by Y .

237 **Proposition 4.1.** *For a covariate vector $\mathbf{X}$, label Y , with causal structure $\mathbf{X} \leftrightarrow Y$, i.e., some*
 238 *covariates cause Y and some covariates are caused by Y , environment variable E , a feature*
 239 *extractor ϕ , and prediction $\hat{Y}$, the predictive information $I(Y; \hat{Y})$ is decomposed as follows:*

$$\begin{aligned}
 I(Y; \hat{Y}) = & \overbrace{I(\phi(\mathbf{X}); Y|E)}^{\text{Cond. informativ.}} - \overbrace{\frac{I(\phi(\mathbf{X}); E|Y)}{2}}^{\text{Variation}} + \overbrace{\frac{I(Y; E)}{2}}^{\text{Label shift}} + \overbrace{\frac{I(\phi(\mathbf{X}); E)}{2}}^{\text{Feature shift}} - \overbrace{\frac{I(Y; E|\phi(\mathbf{X}))}{2}}^{\text{Concept shift}} - \overbrace{I(\phi(\mathbf{X}); Y|\hat{Y})}^{\text{Residual}} \\
 & (1)
 \end{aligned}$$

240 where $I(\phi(\mathbf{X}); Y|\hat{Y})$ is the residual information in $\phi(\mathbf{X})$ for inferring Y that is not captured by the
 241 prediction $\hat{Y}$. The decomposition above also holds when $\phi(\mathbf{X})$ is replaced with $\mathbf{X}$.

Proofs appear in Appendix § A. Equation (1) makes explicit which information terms existing objectives encourage or suppress. IRM [6], for example, constrains a fixed classifier f to remain optimal across environments, which corresponds to increasing $-I(\phi(\mathbf{X}); E|Y)$: if $\phi(\mathbf{X})$ retains environment-specific information after conditioning on Y , a single classifier need not be simultaneously optimal. DANN [23] and the independence criterion for fair classification [22] reduce environment information in the representation by minimizing $I(\phi(\mathbf{X}); E)$. Domain-adaptation methods likewise learn feature properties that are invariant to domains [82, 90]. Although reducing $I(\phi(\mathbf{X}); E)$ can support fairness, it can also reduce predictive performance [98, 33, 22]; Equation (1) shows why, since $I(\phi(\mathbf{X}); E)$ enters the predictive-information expression with a positive coefficient. CDAN [51] instead targets joint alignment, $\mathbb{P}^e(\phi(\mathbf{X}), Y) = \mathbb{P}^{e'}(\phi(\mathbf{X}), Y)$. This makes E independent of $(\phi(\mathbf{X}), Y)$ and implies $\mathbb{P}(Y | \phi(\mathbf{X}), E) = \mathbb{P}(Y | \phi(\mathbf{X})) \implies I(Y; E | \phi(\mathbf{X})) = 0$. Thus, even when $I(Y; E | \phi(\mathbf{X}))$ is not explicitly optimized, joint-distribution alignment drives concept shift toward zero. The label-shift term $I(Y; E)$ is fixed for a given environment distribution, although robust methods can still address shifts in label frequencies [69, 95].

Equation (1) is intentionally general and does not yet specify how U acts on $\mathbf{X}$ and Y . Once that causal role is specified, the appropriate treatment of the terms becomes sharper. If $U \rightarrow Y, U \not\rightarrow \mathbf{X}, Y \rightarrow \mathbf{X}$, then $I(\mathbf{X}; E | Y) = 0$, making the reduction of $I(\phi(\mathbf{X}); E | Y)$ a principled target. If instead $U \not\rightarrow Y, U \rightarrow \mathbf{X}, \mathbf{X} \rightarrow Y$, then $I(Y; E | \mathbf{X}) = 0$, which motivates minimizing $I(Y; E | \phi(\mathbf{X}))$. The hidden-confounding case $U \rightarrow Y$ and $U \rightarrow \mathbf{X}$ is more consequential: the decomposition terms interact rather than separating into a single standard shift assumption. The next subsection resolves this interaction.

4.2 Predictive information under hidden confounding shift

To understand how hidden confounding variable U that cause both $\mathbf{X}$ and Y , impact the predictive information, we consider *two special cases* of the relationship between $\mathbf{X}$ and Y : (i) $\mathbf{X} \rightarrow Y$ and (ii) $Y \rightarrow \mathbf{X}$. When $\mathbf{X} \rightarrow Y$, we obtain the inequalities in (2) because conditioning on Y opens the path $\phi(\mathbf{X}) \leftarrow \mathbf{X} \rightarrow Y \leftarrow U \leftarrow E$ from $\phi(\mathbf{X})$ to E at the collider node Y which results in additional information flow from $\phi(\mathbf{X})$ to E via Y . In contrast, conditioning on $\phi(\mathbf{X})$ partially blocks the information flow from Y to E at the node $\mathbf{X}$ as long as $\phi(\mathbf{X})$ encodes some information about $\mathbf{X}$.

$$(i) \overbrace{I(\phi(\mathbf{X}); E|Y)}^{\text{Variation}} \geq \overbrace{I(\phi(\mathbf{X}); E)}^{\text{Feature shift}} \quad (ii) \overbrace{I(Y; E)}^{\text{Label shift}} \geq \overbrace{I(Y; E|\phi(\mathbf{X}))}^{\text{Concept shift}} \quad (2)$$

Similarly, when $Y \rightarrow \mathbf{X}$, we obtain the inequalities in (3) because conditioning on Y blocks the information flow from $\phi(\mathbf{X})$ to E at the node Y and conditioning on $\phi(\mathbf{X})$ opens the path $Y \rightarrow \mathbf{X} \leftarrow U \leftarrow E$ from Y to E at the node $\mathbf{X}$ because $\phi(\mathbf{X})$ is the child of $\mathbf{X}$ and conditioning on the child of a collider opens a path via that collider [58] (recall the preliminaries in § 4.1).

$$(i) \overbrace{I(\phi(\mathbf{X}); E)}^{\text{Feature shift}} \geq \overbrace{I(\phi(\mathbf{X}); E|Y)}^{\text{Variation}} \quad (ii) \overbrace{I(Y; E|\phi(\mathbf{X}))}^{\text{Concept shift}} \geq \overbrace{I(Y; E)}^{\text{Label shift}} \quad (3)$$

Using the inequalities in (2), (3), we refine the predictive information decomposition as follows.

Proposition 4.2. For a covariate vector $\mathbf{X}$, label Y , an environment variable E , a feature extractor ϕ , the prediction $\hat{Y}$, and an unobserved confounding variable U that cause both $\mathbf{X}$ and Y , and if either (i) $\mathbf{X} \rightarrow Y$ or (ii) $Y \rightarrow \mathbf{X}$, the predictive information $I(Y; \hat{Y})$ can be decomposed as follows:

$$I(Y; \hat{Y}) = \overbrace{I(\phi(\mathbf{X}); Y|E)}^{\text{Cond. informativeness}} - \overbrace{I(\phi(\mathbf{X}); Y|\hat{Y})}^{\text{Residual}} \quad (4)$$

In Proposition 4.2, we tackle the two cases: $\mathbf{X} \rightarrow Y$ and $Y \rightarrow \mathbf{X}$ separately. However, the causal relationship $\mathbf{X} \rightarrow Y$ is more prevalent in real-world tabular prediction tasks. For instance, 11 out of 16 datasets considered in recent benchmark studies [57, 50, 24] follow the causal structure $\mathbf{X} \rightarrow Y$. In the remaining 5 out of 16 datasets, the number of covariates that are caused by Y is much smaller than the number of covariates that cause Y . We present the key takeaway from Proposition 4.2 below.

Compared to the decomposition in (1), the decomposition in (4) (which is obtained for the special cases: (i) $U \rightarrow \mathbf{X}, U \rightarrow Y, \mathbf{X} \rightarrow Y$, (ii) $U \rightarrow \mathbf{X}, U \rightarrow Y, Y \rightarrow \mathbf{X}$) is free from the terms: *variation, label shift, feature shift, and concept shift*. That is, under hidden confounding shift, when either $\mathbf{X} \rightarrow Y$ or $Y \rightarrow \mathbf{X}$, the predictive information can be maximized by maximizing the difference: *conditional informativeness* – *residual*.

284 **Maximizing conditional informativeness:** In a very recent work, Prashant et al. [62] proposed a
 285 mixture of experts (MoE) model for OOD generalization under hidden confounding shift, where
 286 each expert corresponds to a specific hidden confounder assignment. That is, each expert focuses on
 287 maximizing the performance within the environment corresponding to a particular value of the hidden
 288 confounder. Thus, learning MoE models aligns with the goal of maximizing the conditional infor-
 289 mativeness $I(\phi(\mathbf{X}); Y|E)$. Equation (4) provides theoretical justification for the use of MoE-type
 290 models. Beyond supporting such models, the Equation (4) also highlights the need for methods that
 291 operate under more general confounding shift settings. For instance, the method proposed by Prashant
 292 et al. [62] assumes overlapping confounding support i.e., $\text{supp}(\mathbb{P}(U|\mathcal{E}_{te})) \subseteq \text{supp}(\mathbb{P}(U|\mathcal{E}_{tr}))$, and
 293 the proxy variables to be discrete-valued.

The results in Figure 2 illustrate that, in a simple setting without confounding overlap and with a continuous-valued proxy, a linear regression model can successfully recover the correct causal relationship between observed covariates and the target across environments when provided with environment-specific information. This serves as a motivational example that emphasizes the need for OOD generalization methods that remain effective even when confounding support is limited and the confounder is continuous-valued.

294

295 **Minimizing residual:** The residual term $I(\phi(\mathbf{X}); Y | \hat{Y})$ quantifies how much additional information
 296 $\phi(\mathbf{X})$ provides about the true label Y beyond what is already contained in the prediction $\hat{Y}$. The
 297 residual term can be expressed as: $I(\phi(\mathbf{X}); Y | \hat{Y}) = H(Y | \hat{Y}) - H(Y | \phi(\mathbf{X}), \hat{Y})$. The conditional
 298 entropy $H(Y | \hat{Y})$ is related to the expected cross-entropy loss as: $\mathbb{E}_{Y, \hat{Y}}[\ell_{\text{CE}}(Y, \hat{Y})] = H(Y |$
 299 $\hat{Y}) + \text{calibration error}$ [12, 10, 29]. Here $H(Y | \hat{Y})$ is known as the *refinement error* that measures
 300 the model’s ability to distinguish between classes. Minimizing the residual term $I(\phi(\mathbf{X}); Y | \hat{Y})$ thus
 301 involves two complementary strategies. First, one can directly minimize $I(\phi(\mathbf{X}); Y | \hat{Y})$ by making
 302 $\phi(\mathbf{X})$ conditionally independent of the target Y given $\hat{Y}$, i.e., $Y \perp\!\!\!\perp \phi(\mathbf{X}) | \hat{Y}$. Alternatively, one can
 303 minimize the refinement error $H(Y | \hat{Y})$, for instance, through training techniques that minimize
 304 refinement error for accurate predictions before minimizing the calibration error [10].

305 4.3 Impact of additional informative variables

306 In § 4.2, we present the optimal strategies for generalization when the confounder is unobserved.
 307 In practice, we sometimes have access to proxies for the hidden confounding variable [4, 85, 62],
 308 which can be leveraged to substitute for the hidden confounding variable. We now study the impact
 309 of adding such non-causal but informative covariates on predictive information.

310 **Definition 4.3** (Informative Covariates). *A set of covariates $\mathbf{X}_I$ that are not causally related to Y i.e.,*
 311 *neither ancestors nor descendants of Y are in $\mathbf{X}_I$, are said to be informative to Y if $\mathbf{X}_I$ and Y are*
 312 *not independent of each other given other causally related covariates $\mathbf{X}$ and E i.e., $Y \not\perp\!\!\!\perp \mathbf{X}_I | \mathbf{X}, E$.*

313 Since $U \rightarrow Y$, any covariate that is informative to U , is also informative to Y and vice-versa. From
 314 the predictive information decomposition in (1), to maximize $I(Y; \hat{Y})$, it is required to minimize the
 315 concept shift $I(Y; E | \phi(\mathbf{X}))$. Liu et al. [50] suggest based on their empirical analysis that collecting
 316 additional covariates $\mathbf{X}_I$ such that $\mathbb{P}(Y|\mathbf{X}, \mathbf{X}_I)$ is more stable across environments i.e., reducing
 317 concept shift, improves the OOD test accuracy. We theoretically show that utilizing more informative
 318 variables helps maximizing predictive information by maximizing or minimizing certain terms in (1).

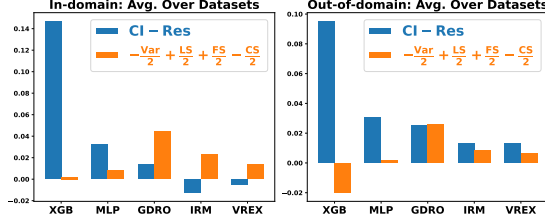
319 **Proposition 4.3.** *If $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and if $\mathbf{X}_I$ are informative variables such that*
 320 *$Y \not\perp\!\!\!\perp \mathbf{X}_I | \mathbf{X}, E$, then we have the following inequalities:*

$$\begin{aligned} (i) & I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); Y|E) > I(\phi_1(\mathbf{X}); Y|E) & (ii) & I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E) > I(\phi_1(\mathbf{X}); E) \\ (iii) & I(Y; E|\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\})) < I(Y; E|\phi_1(\mathbf{X})) & (iv) & I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E | Y) > I(\phi_1(\mathbf{X}); E | Y) \end{aligned} \quad (5)$$

321 Proposition 4.3 shows that adding informative covariates increases conditional informativeness (5.i)
 322 and feature shift (5.ii), while reducing concept shift (5.iii). However, adding informative covariates
 323 also amplifies variation (5.iv), which may necessitate dedicated strategies to control the variation.

Table 2: Quantifying distribution shifts. Mean $\pm$ standard deviation is computed over 10 random subsets of 40,000 samples. Statistical significance against a mean of zero is assessed via one-sample t-tests ($\alpha = 0.05$), confirming all measures are significantly different from zero (p-value ≈ 0).

Dataset	Conditional Covariate Shift $I(\mathbf{X}; E Y)$	Label shift $I(Y; E)$	Covariate Shift $I(\mathbf{X}; E)$	Concept shift $I(Y; E \mathbf{X})$
Readmission	0.107 ± 0.002	0.068 ± 0.002	0.097 ± 0.002	2.032 ± 0.000
Food stamps	0.126 ± 0.004	0.030 ± 0.003	0.108 ± 0.001	2.118 ± 0.001
Income	0.168 ± 0.002	0.075 ± 0.003	0.147 ± 0.001	2.059 ± 0.002
Public coverage	0.231 ± 0.002	0.412 ± 0.006	0.222 ± 0.002	1.945 ± 0.001
Unemployment	0.117 ± 0.001	0.019 ± 0.002	0.114 ± 0.002	2.010 ± 0.003
Diabetes	0.032 ± 0.002	0.048 ± 0.001	0.022 ± 0.002	2.132 ± 0.001
Hypertension	0.090 ± 0.002	0.183 ± 0.003	0.037 ± 0.001	1.883 ± 0.004
ASSISTments	0.293 ± 0.002	0.260 ± 0.001	0.306 ± 0.002	0.367 ± 0.004



Method	Avg. ID Test Acc.	Avg. OOD Test Acc.
XGB	85.22	77.00
MLP	83.64	73.29
GDRO	78.40	72.28
IRM	71.17	64.89
VREX	59.74	63.06

Figure 4: The difference *conditional informativeness* – *residual* in the plots is positively correlated with the average ID and OOD test accuracy over the eight datasets shown in the table on the right.

5 Experimental results

We conduct experiments on both real-world and synthetic datasets to analyze: (i) the presence of hidden confounding shift in real-world data, (ii) how the components of the decomposition in (1) affect OOD generalization under hidden confounding shifts, and (iii) the role of informative covariates in improving generalization. We consider eight real-world tabular benchmark datasets: *Food stamps*, *Readmission*, *Income*, *Public coverage*, *Unemployment*, *Diabetes*, *Hypertension*, and *ASSISTments*. These datasets and corresponding domain splits are adopted from *TableShift* benchmark [24]. We use ID test and OOD test accuracies to measure the performance of models. We perform experiments on two ERM-based methods: XGBoost (XGB) [15] and multi-layer perceptron (MLP), two domain generalization methods: IRM [6], VREX [38], and one robust learning based method: Group DRO (GDRO) [69]. We present average results over 5 different random hyperparameter choices as described in Appendix B. The effect of different random seeds is statistically insignificant for these datasets [24]. Additional details of the experimental setup are provided in Appendix § B. Results with hyperparameter tuning are presented in Appendix C.

Hidden confounding shift: Recall that hidden confounding shifts induce observable shifts in the distributions $\mathbb{P}(\mathbf{X})$, $\mathbb{P}(\mathbf{X} | Y)$, $\mathbb{P}(Y)$, and $\mathbb{P}(Y | \mathbf{X})$. Results in Table 2 show that label shift, covariate shift, conditional covariate shift, and concept shift are all present in real-world datasets, indicating the presence of hidden confounding shifts. For the qualitative analysis, we query GPT-4o [1] to list potential hidden confounders for several benchmark datasets [24, 57]. These results are presented in Appendix § E. For instance, in the *food stamps* dataset, unmeasured factors such as *economic policies* specific to each state may influence both *household income* and *food stamp reciprocity*.

Conditional informativeness vs. accuracy: From Proposition 4.2, under hidden confounding shift, maximizing the difference *conditional informativeness* – *residual* is essential for maximizing predictive information. Results on the first six datasets (that have more than one training environments [57, 24]) listed in Table 2 are shown in Figure 4. Dataset-wise results are in Appendix C. These results show that the difference *conditional informativeness* – *residual* is positively correlated with ID and OOD accuracy, with XGB being the best performing model with highest value of *conditional informativeness* – *residual*. As shown in the dataset-wise results in Appendix C, we observe that in many settings, due to hidden confounding shifts, the bars denoted with orange are close to zero because the sum $-\text{variation} / 2 + \text{label shift} / 2 + \text{feature shift} / 2 - \text{concept shift} / 2$ is zero under hidden confounding shift (Proposition 4.2). The relative ID / OOD performance of the methods shown in Figure 1 is consistent with previously reported results in the literature [57, 24].

Table 3: For all methods, sign consistency value is high for concept shift (CS). However, the sign consistency metric for conditional informativeness (CI) is crucial for generalization according to Proposition 4.2, and XGB excels at this. Res: residual, Var: variation, FS: feature shift, C: causal, AC: arguably causal, A: all.

Method	Sign Consistency Metric ($\uparrow$)					ID-Test Accuracy ($\uparrow$)			OOD-Test Accuracy ($\uparrow$)		
	CI	Var	FS	CS	Res	C	AC	A	C	AC	A
XGB	0.89	0.61	0.36	0.89	0.17	81.24	84.84	85.22	69.92	76.72	77.00
MLP	0.75	0.64	0.28	0.89	0.33	80.51	81.93	83.64	69.70	73.84	73.29
GDRO	0.69	0.67	0.19	0.94	0.36	80.92	83.76	78.40	69.61	73.49	72.28
IRM	0.78	0.61	0.25	0.92	0.25	67.91	68.22	71.17	61.21	62.96	64.89
VREX	0.64	0.61	0.25	0.89	0.44	54.39	62.32	59.74	56.62	63.53	63.06

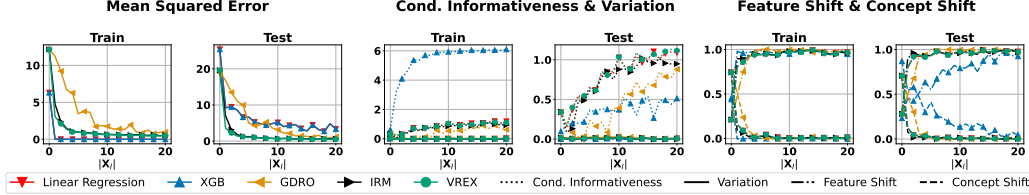


Figure 5: Adding more proxy variables $\mathbf{X}_I$ of U that are informative to Y helps in reducing MSE, increasing conditional informativeness and feature shift while reducing concept shift.

Informative covariates vs. accuracy: We now study how inclusion of informative covariates helps in achieving better OOD generalization. To this end, we adopt the covariate partitioning from [57], where all available covariates are grouped into three nested covariate subsets: *causal covariates* $\subseteq$ *arguably causal covariates* $\subseteq$ *all covariates* (see Appendix § B). We evaluate how the terms in the decomposition in (1) affect ID and OOD performance when going from one covariate subset to another covariate subset. To this end, we use what we call the *sign consistency metric*, which works as follows. When we move from one covariate set to a larger covariate set, for any term in (1) with a positive coefficient, such as conditional informativeness, we count how often its value rises when accuracy improves. For negatively weighted terms, such as concept shift, we count how often its value decreases when accuracy improves. The metric is the fraction of observations where a term’s change aligns with its *beneficial* direction, thus capturing how reliably it contributes to better generalization. See Appendix § B for the formal definition. Table 3 shows that both conditional informativeness and concept shift exhibit high sign consistency with accuracy gains. While reducing concept shift is crucial, XGB further enhances conditional informativeness compared to other methods, yielding additional performance improvements. Dataset-specific results are presented in Appendix § C.

Note that in real-world datasets, it is often challenging to verify whether *all available covariates* include *every relevant or sufficiently informative* variable. To further test our hypothesis that informative covariates enhance generalization, we conduct experiments on synthetic data with a known causal structure: $U \rightarrow X, U \rightarrow Y, X \rightarrow Y$, and $U \rightarrow \mathbf{X}_I$, where $\mathbf{X}_I$ denotes informative covariates caused by the hidden confounder U . We observe that including $\mathbf{X}_I$ leads to improvements in conditional informativeness and feature shift, while reducing concept shift, as shown in Figure 5. Notably, the variation component remains consistently zero due to the simplicity of the problem. The synthetic data generation process and additional results are provided in Appendix § D.

6 Discussion, limitations, and future work

In summary, we decompose the predictive information between model outputs and true outcomes to reveal the factors limiting OOD generalization (Equation (1)). We argue that hidden confounding shifts—reflected in covariate and outcome distribution changes across environments—are unavoidable, and no finite set of observed covariates or proxies can fully resolve the induced ambiguity. Since confounding entangles covariate, label, concept, and feature shifts (inequalities in (2), (3)), treating these in isolation is ineffective. We show that learning environment-specific input–output mappings is more effective than enforcing a single invariant predictor (Equation (4)). By highlighting the inevitability of hidden confounding and the need to address it directly, our work lays a foundation for more robust OOD methods. Key challenges ahead include: (i) developing algorithms effective under limited confounder overlap, (ii) quantifying the cost–accuracy trade-off of acquiring informative covariates, (iii) handling entangled shifts without relying on untestable proxy assumptions, and (iv) modeling ambiguity from unobserved confounders to inspire new OOD-robust paradigms.

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670 Appendix

671 A Proofs of propositions

672 **Proposition 4.1.** For a covariate vector $\mathbf{X}$, label Y , with causal structure $\mathbf{X} \leftrightarrow Y$, i.e., some
 673 covariates cause Y and some covariates are caused by Y , environment variable E , a feature
 674 extractor ϕ , and prediction $\hat{Y}$, the predictive information $I(Y; \hat{Y})$ is decomposed as follows:

$$I(Y; \hat{Y}) = \overbrace{I(\phi(\mathbf{X}); Y|E)}^{\text{Cond. informativ.}} - \overbrace{\frac{I(\phi(\mathbf{X}); E|Y)}{2}}^{\text{Variation}} + \overbrace{\frac{I(Y; E)}{2}}^{\text{Label shift}} + \overbrace{\frac{I(\phi(\mathbf{X}); E)}{2}}^{\text{Feature shift}} - \overbrace{\frac{I(Y; E|\phi(\mathbf{X}))}{2}}^{\text{Concept shift}} - \overbrace{I(\phi(\mathbf{X}); Y|\hat{Y})}^{\text{Residual}} \quad (1)$$

675 where $I(\phi(\mathbf{X}); Y|\hat{Y})$ is the residual information in $\phi(\mathbf{X})$ for inferring Y that is not captured by the
 676 prediction $\hat{Y}$. The decomposition above also holds when $\phi(\mathbf{X})$ is replaced with $\mathbf{X}$.

677 *Proof.* From the causal graph shown in Figure 3 of the main paper, in the causal substructure:
 678 $\hat{Y} \leftarrow \phi(\mathbf{X}) \leftarrow \mathbf{X} \leftrightarrow Y$, $\hat{Y}$ provides no additional information in predicting Y given either $\phi(\mathbf{X})$ or
 679 $\mathbf{X}$. This is because, the path $\hat{Y} \leftarrow \phi(\mathbf{X}) \leftarrow \mathbf{X} \leftrightarrow Y$, between $\hat{Y}$ and Y is blocked by either $\{\phi(\mathbf{X})\}$
 680 or $\{\mathbf{X}\}$. This implies $\hat{Y} \perp\!\!\!\perp Y|\phi(\mathbf{X})$ and $\hat{Y} \perp\!\!\!\perp Y|\mathbf{X}$. Now consider the expansion of $I(Y; \hat{Y})$.

$$\begin{aligned} I(Y; \hat{Y}) &= I(Y; \phi(\mathbf{X}), \hat{Y}) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Introduce } \phi(\mathbf{X})) \\ &= I(\phi(\mathbf{X}); Y) + \underbrace{I(Y; \hat{Y}|\phi(\mathbf{X}))}_0 - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Expand and use } \hat{Y} \perp\!\!\!\perp Y | \phi(\mathbf{X})) \\ &= I(\phi(\mathbf{X}), E; Y) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Introduce } \mathbf{E}) \\ &= I(Y; E) + I(\phi(\mathbf{X}); Y|E) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Expand}) \end{aligned} \quad (6)$$

681 Similarly,

$$\begin{aligned} I(Y; \hat{Y}) &= I(Y; \phi(\mathbf{X}), \hat{Y}) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Introduce } \phi(\mathbf{X})) \\ &= I(\phi(\mathbf{X}); Y) + \underbrace{I(Y; \hat{Y}|\phi(\mathbf{X}))}_0 - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Expand and use } \hat{Y} \perp\!\!\!\perp Y | \phi(\mathbf{X})) \\ &= I(Y, E; \phi(\mathbf{X})) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Introduce } \mathbf{E}) \\ &= I(\phi(\mathbf{X}); E) + I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Expand}) \end{aligned} \quad (7)$$

682 Summing Equations (6) and (7) and grouping the terms gives the desired expression:

$$\begin{aligned} I(Y; \hat{Y}) &= I(\phi(\mathbf{X}); Y|E) - \frac{I(\phi(\mathbf{X}); E|Y)}{2} + \frac{I(Y; E)}{2} \\ &\quad + \frac{I(\phi(\mathbf{X}); E)}{2} - \frac{I(Y; E|\phi(\mathbf{X}))}{2} - I(\phi(\mathbf{X}); Y|\hat{Y}). \end{aligned}$$

683 To get the similar decomposition with $\mathbf{X}$ instead of $\phi(\mathbf{X})$, use $\hat{Y} \perp\!\!\!\perp Y | \mathbf{X}$ and introduce $\mathbf{X}$ into the
 684 expansion instead of $\phi(\mathbf{X})$. $\square$

685 **Proposition 4.2.** For a covariate vector $\mathbf{X}$, label Y , an environment variable E , a feature extractor
 686 ϕ , the prediction $\hat{Y}$, and an unobserved confounding variable U that cause both $\mathbf{X}$ and Y , and if
 687 either (i) $\mathbf{X} \rightarrow Y$ or (ii) $Y \rightarrow \mathbf{X}$, the predictive information $I(Y; \hat{Y})$ can be decomposed as follows:

$$I(Y; \hat{Y}) = \overbrace{I(\phi(\mathbf{X}); Y|E)}^{\text{Cond. informativeness}} - \overbrace{I(\phi(\mathbf{X}); Y|\hat{Y})}^{\text{Residual}} \quad (4)$$

688 *Proof.* Following the causal graph shown in Figure 3 of the main paper, we prove the proposition in
 689 two separate cases: $\mathbf{X} \rightarrow Y$ and $Y \rightarrow \mathbf{X}$.

690 **Case 1: $\mathbf{X} \rightarrow Y$.** In this case, we have the following inequality because conditioning on Y opens the
 691 path $\phi(\mathbf{X}) \leftarrow \mathbf{X} \rightarrow Y \leftarrow U \leftarrow E$ between $\phi(\mathbf{X})$ and E at the collider node Y :

$$I(\phi(\mathbf{X}); E|Y) \geq I(\phi(\mathbf{X}); E). \quad (8)$$

692 Additionally, considering the causal substructure: $Y \leftarrow \mathbf{X} \leftarrow U \leftrightarrow E$, we have the following
 693 inequality because conditioning on $\phi(\mathbf{X})$ partially blocks the information flow from Y to E at the
 694 node $\mathbf{X}$ as long as $\phi(\mathbf{X})$ encodes some information about $\mathbf{X}$:

$$I(Y; E) \geq I(Y; E|\phi(\mathbf{X})). \quad (9)$$

695 Now consider the following expansion from Equation (7) in the proof of Proposition 4.1.

$$\begin{aligned} I(Y; \hat{Y}) &= I(\phi(\mathbf{X}); E) + I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Equation (7)}) \\ &\leq I(\phi(\mathbf{X}); Y|E) + I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Using (8)}) \\ &= I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \end{aligned}$$

696 Similarly, consider the following expansion from Equation (6) from the proof of Proposition 4.1:

$$\begin{aligned} I(Y; \hat{Y}) &= I(Y; E) + I(\phi(\mathbf{X}); Y|E) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Equation (6)}) \\ &\geq I(Y; E) + I(\phi(\mathbf{X}); Y|E) - I(Y; E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Using (9)}) \\ &= I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \end{aligned}$$

697 That is, we have $I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \leq I(Y; \hat{Y}) \leq I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y})$.

698 **Case 2: $Y \rightarrow \mathbf{X}$.** In this scenario, we have the following inequality because conditioning on $\phi(\mathbf{X})$
 699 opens the path $E \rightarrow U \rightarrow \mathbf{X} \leftarrow Y$ between Y and E at the collider node $\mathbf{X}$ because $\phi(\mathbf{X})$ is a child
 700 of $\mathbf{X}$ [58].

$$I(Y; E|\phi(\mathbf{X})) \geq I(Y; E) \quad (10)$$

701 Additionally, we have the following inequality because conditioning on Y blocks the path $\phi(\mathbf{X}) \leftarrow$
 702 $\mathbf{X} \leftarrow Y \leftarrow U \leftrightarrow E$ at Y and hence less information flow between $\phi(\mathbf{X})$ and E .

$$I(\phi(\mathbf{X}); E) \geq I(\phi(\mathbf{X}); E|Y) \quad (11)$$

703 Now consider the following expansion from Equation (6) from the proof of Proposition 4.1.

$$\begin{aligned} I(Y; \hat{Y}) &= I(Y; E) + I(\phi(\mathbf{X}); Y|E) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Equation (6)}) \\ &\leq I(\phi(\mathbf{X}); Y|E) + I(Y; E|\phi(\mathbf{X})) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Using (10)}) \\ &= I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \end{aligned}$$

704 Similarly, consider the following expansion from Equation (7) from the proof of Proposition 4.1.

$$\begin{aligned} I(Y; \hat{Y}) &= I(\phi(\mathbf{X}); E) + I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Equation (7)}) \\ &\geq I(\phi(\mathbf{X}); Y|E) + I(\phi(\mathbf{X}); E) - I(\phi(\mathbf{X}); E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Using (11)}) \\ &= I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \end{aligned}$$

705 That is, we have $I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \leq I(Y; \hat{Y}) \leq I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y})$.

706 Since the upper bound and lower bounds are the same in both cases, we have the desired result
 707 $I(Y; \hat{Y}) = I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y})$. $\square$

708 **Proposition 4.3.** If $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and if $\mathbf{X}_I$ are informative variables such that
 709 $Y \not\perp \mathbf{X}_I \mid \mathbf{X}, E$, then we have the following inequalities:

$$\begin{aligned} (i) & I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); Y|E) > I(\phi_1(\mathbf{X}); Y|E) \quad (ii) I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E) > I(\phi_1(\mathbf{X}); E) \\ (iii) & I(Y; E|\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\})) < I(Y; E|\phi_1(\mathbf{X})) \quad (iv) I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E \mid Y) > I(\phi_1(\mathbf{X}); E \mid Y) \end{aligned} \quad (5)$$

710 *Proof.* (i) By the chain rule of mutual information, we have: $I(\mathbf{X}, \mathbf{X}_I; Y|E) = I(\mathbf{X}; Y|E) +$
711 $I(\mathbf{X}_I; Y|E, \mathbf{X})$. We have $I(\mathbf{X}_I; Y|E, \mathbf{X}) > 0$ because $Y \not\perp\!\!\!\perp \mathbf{X}_I \mid \mathbf{X}, E$. Hence, we have
712 $I(\mathbf{X}, \mathbf{X}_I; Y|E) > I(\mathbf{X}; Y|E)$. Since $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and mutual information is
713 invariant to invertible transformations, we have $I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); Y|E) > I(\phi_1(\mathbf{X}); Y|E)$.

714 (ii) We consider the causal graph shown in Figure 3. From the chain rule of mutual information,
715 we have: $I(\mathbf{X}, \mathbf{X}_I; E) = I(\mathbf{X}; E) + I(\mathbf{X}_I; E|\mathbf{X})$. Since $E \leftrightarrow U \rightarrow Y$, any covariate that is
716 informative to Y is also informative to U . Since $E \leftrightarrow U$, we have $I(\mathbf{X}_I; E|\mathbf{X}) > 0$, and hence
717 $I(\mathbf{X}, \mathbf{X}_I; E) > I(\mathbf{X}; E)$. Since $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and mutual information is
718 invariant to invertible transformations, we have $I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E) > I(\phi_1(\mathbf{X}); E)$.

719 (iii) Consider $I(Y; E, \mathbf{X}_I | \mathbf{X})$. It can be expressed in two ways:

$$I(Y; E, \mathbf{X}_I | \mathbf{X}) = I(Y; E | \mathbf{X}) + I(Y; \mathbf{X}_I | E, \mathbf{X}) = I(Y; \mathbf{X}_I | \mathbf{X}) + I(Y; E | \mathbf{X}_I, \mathbf{X})$$

720 Equating and rearranging the terms gives the following.

$$I(Y; E | \mathbf{X}) - I(Y; E | \mathbf{X}_I, \mathbf{X}) = I(Y; \mathbf{X}_I | \mathbf{X}) - I(Y; \mathbf{X}_I | E, \mathbf{X})$$

721 Since $Y \not\perp\!\!\!\perp \mathbf{X}_I \mid E, \mathbf{X}$, we have $I(Y; \mathbf{X}_I | \mathbf{X}) > I(Y; \mathbf{X}_I | E, \mathbf{X})$ because conditioning usually
722 reduces mutual information unless the additional conditioning variable opens any collider paths. Here
723 E is neither a collider nor a descendant of any collider. Thus, $I(Y; E | \mathbf{X}) > I(Y; E | \mathbf{X}_I, \mathbf{X})$. Since
724 $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and mutual information is invariant to invertible transformations,
725 we have $I(Y; E | \phi_1(\mathbf{X})) > I(Y; E | \phi_2(\{\mathbf{X}_I \cup \mathbf{X}\}))$.

726 (iv) By the chain rule of mutual information, we have: $I(\mathbf{X}, \mathbf{X}_I; E | Y) = I(\mathbf{X}; E | Y) +$
727 $I(\mathbf{X}_I; E|\mathbf{X}, Y)$. Since $E \leftrightarrow U \rightarrow Y$, any covariate that is informative to Y is also informative to
728 U . Since $E \leftrightarrow U$, we have $I(\mathbf{X}_I; E|\mathbf{X}, Y) > 0$, and hence $I(\mathbf{X}, \mathbf{X}_I; E | Y) > I(\mathbf{X}; E | Y)$. Since
729 $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and mutual information is invariant to invertible transformations,
730 we have $I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E | Y) > I(\phi_1(\mathbf{X}); E | Y)$. $\square$

731 B Experimental setup

732 In this section, we detail our experimental setup. The impact of random seeds on the results is
733 statistically insignificant in many settings [24]. For the results in the main paper, we report the mean
734 and standard deviation across five random hyperparameter values (sampled from their respective
735 domains in Table B1) for each dataset-method combination. Since we evaluate models on ID test and
736 OOD test data, to evaluate mutual information terms, we use the nonparametric entropy estimation
737 toolbox [37, 77, 78]. Following [24], we evaluate mutual information using both in-distribution (ID)
738 test data (from training domains) and out-of-distribution (OOD) test data (from test domains).

739 For consistency, we select 20,000 random samples
740 from each ID and OOD test data, though
741 some datasets contain fewer than 20,000 sam-
742 ples. Throughout our experiments, the sample
743 size has a relatively insignificant effect on
744 the evaluated mutual information terms. For
745 instance, as shown in Figure B1, the terms in
746 Equation (1) computed for the representations
747 of the XGB model on the *income* dataset stayed
748 relatively consistent as we vary sample size. Fea-
749 tures $\phi(\mathbf{X})$ are extracted from the layer preced-
750 ing the classification head; for XGBoost, we
751 combine output margins and SHAP values from the model. All experiments were conducted on a
752 single NVIDIA RTX A6000 GPU.

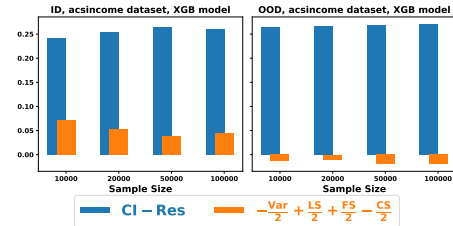


Figure B1: Evaluated terms of predictive information are relatively consistent across various sample sizes on the *income* dataset.

753 In our experiments studying the impact of informative covariates on predictive information, we follow
754 the categorization introduced by Nastl and Hardt [57], partitioning covariates into: *causal*, *arguably*
755 *causal*, and *all covariates*. Causal covariates have clearly established influence on the target, with well-
756 supported directionality and unlikely reverse causation. However, strict reliance on causal variables
757 risks omitting relevant parents due to knowledge gaps. Arguably causal covariates have uncertain
758 causal relationships, meeting at least one of: (1) being known causal, (2) having plausible but poten-
759 tially bidirectional influence, or (3) likely (but unconfirmed) causal effect. These groups approximate

Table B1: Hyperparameters and their possible values considered in this paper.

Model	Hyperparameter	Search values
XGBoost	Learning rate	[0.1, 0.2, 0.3]
	Maximum tree depth	[4, 5, 6]
	Minimum child weight	[0.1, 1, 10]
	Gamma (min. loss reduction)	[0.0001, 0.001, 0.01]
	Subsample ratio	[0.5, 0.6, 0.7]
	Column subsample per tree	[0.5, 0.6, 0.7]
	L1 regularization (α)	[0.0001, 0.001, 0.01]
	L2 regularization (λ)	[0.0001, 0.001, 0.01]
MLP	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
	Number of epochs	[1, 2, 3]
GroupDRO	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	Group-weights step size	[0.01, 0.05, 0.1]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
IRM	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	Dropout rate	[0.0, 0.1, 0.2]
	IRM penalty weight (λ)	[0.01, 0.05, 0.1]
	IRM penalty anneal iterations	[1, 2, 3]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
VREX	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	VREX penalty anneal iterations	[1, 2, 3]
	VREX penalty weight (λ)	[0.1, 10, 100]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
VREX	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	VREX penalty anneal iterations	[1, 2, 3]
	VREX penalty weight (λ)	[0.1, 10, 100]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
VREX	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	VREX penalty anneal iterations	[1, 2, 3]
	VREX penalty weight (λ)	[0.1, 10, 100]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]

760 true causal parents based on available knowledge, though relationships may be confounded. Some
761 datasets additionally contain *anti-causal* covariates where the target likely affects them but not vice
762 versa. The *all covariates* set includes all observed variables regardless of causal status. The sign con-
763 sistency metric introduced in the main paper § 5 is defined as follows. For each $(m, \sigma_m) \in M$ where
764 $M = \{(\text{conditional informativeness}, +1), (\text{variation}, -1), (\text{label shift}, +1), (\text{feature shift}, +1),$
765 $(\text{concept shift}, -1), \text{ and } (\text{residual}, -1)\}$, for a specific dataset, we compute:

$$C_m = \frac{1}{2|P|} \sum_{(s,t) \in P} [\mathbb{I}(\sigma_m \cdot (\mu_t^{\text{tr}}(m) - \mu_s^{\text{tr}}(m)) > 0) + \mathbb{I}(\sigma_m \cdot (\mu_t^{\text{te}}(m) - \mu_s^{\text{te}}(m)) > 0)]$$

766 Where P is the set of covariate set pairs (s, t) with $t \supset s$. For example, s can be causal covariates
767 and t can be arguably causal covariates. There is a factor of 2 in $2P$ that accounts for both train and

test data. $\mu_s^{tr/te}(m)$ is the mean of measure m for covariate set s in training/testing data. $\mathbb{I}(\cdot)$ is the indicator function. $|P| = 3$ for covariate settings $\{causal, arguably\ causal, all\}$. To get final sign consistency metric value for each m , we average across datasets: $\bar{C}_m = \frac{1}{|D|} \sum_{d \in D} C_m^{(d)}$ where $|D|$ is the number of datasets.

C Additional results on real-world datasets

In this section, we present additional results on real-world datasets. Figures C2, C3 show the comparison of the terms: *conditional informativeness* – *residual* and *–variation / 2 + label shift / 2 + feature shift / 2 – concept shift / 2* for each dataset (their aggregation is in the main paper Figure 4 Left). The key takeaway from these results is that, from Proposition 4.1, the sum of blue and orange terms is positively correlated with the overall model performance. At the same time, due to potential hidden confounding shift in real-world data, the contribution of the difference *conditional informativeness* – *residual* is higher towards the overall predictive information when compared to the contribution of *–variation / 2 + label shift / 2 + feature shift / 2 – concept shift / 2*. This serves as a motivation to maximize conditional informativeness for better OOD generalization under a hidden confounding shift.

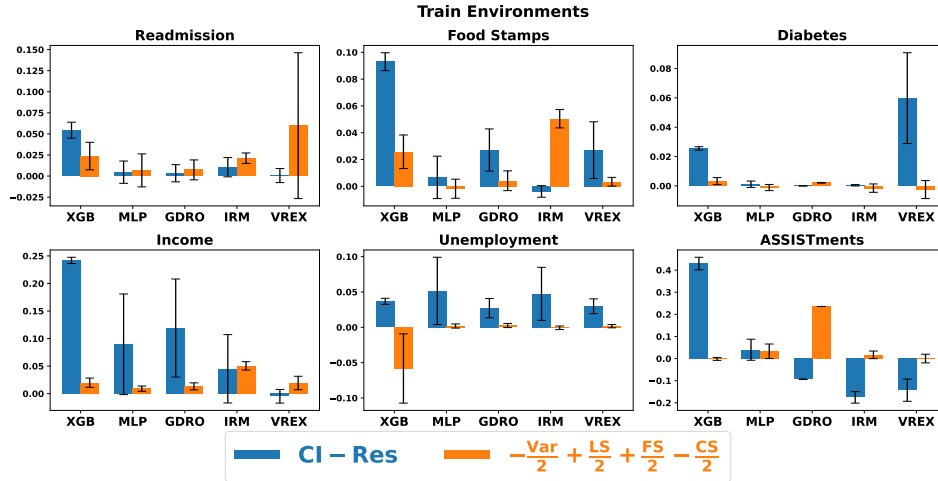


Figure C2: Dataset specific results on ID data.

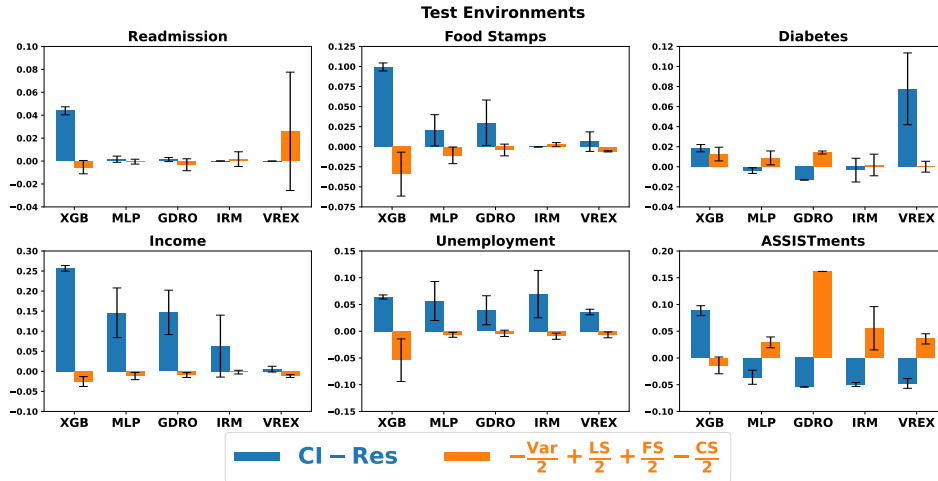


Figure C3: Dataset-specific results on OOD data.

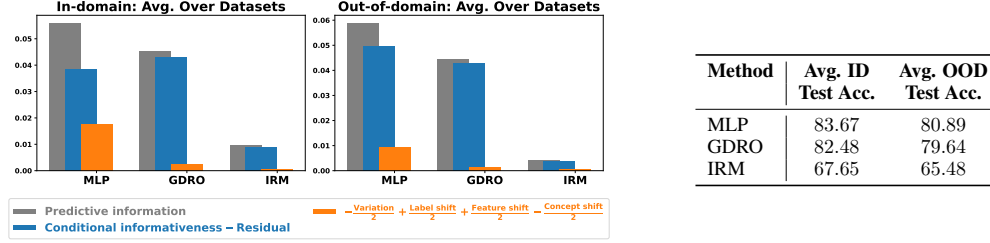


Figure C4: The difference *conditional informativeness* – *residual* in the plots is positively correlated with the average ID and OOD test accuracy over five datasets shown in the table on the right. The contribution of the difference *conditional informativeness* – *residual* is higher towards the overall predictive information when compared to the contribution of $-\text{variation} / 2 + \text{label shift} / 2 + \text{feature shift} / 2 - \text{concept shift} / 2$.

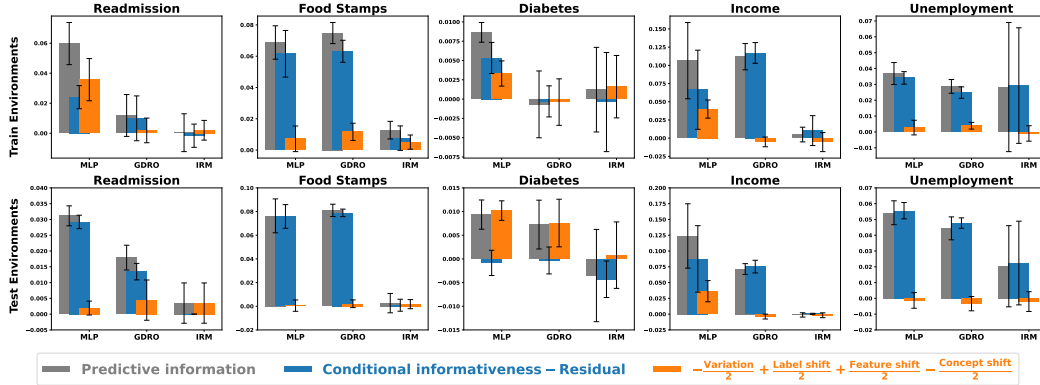


Figure C5: Dataset specific results. Comparison of the terms in predictive information.

Table C2: Comparison of ID and OOD test accuracies of models.

Method	Readmission		Food stamps		Diabetes	
	ID test acc.	OOD test acc.	ID test acc.	OOD test acc.	ID test acc.	OOD test acc.
MLP	65.78 ± 0.14	61.75 ± 0.22	84.74 ± 0.03	82.00 ± 0.06	87.66 ± 0.02	83.22 ± 0.05
GDRO	60.47 ± 1.08	57.60 ± 0.42	84.47 ± 0.04	81.41 ± 0.09	87.47 ± 0.10	82.90 ± 0.10
IRM	50.28 ± 7.35	50.85 ± 1.74	80.91 ± 0.00	78.01 ± 0.00	42.54 ± 36.5	43.48 ± 31.9
	Income		Unemployment			
	ID test acc.	OOD test acc.	ID test acc.	OOD test acc.		
MLP	82.92 ± 0.05	81.46 ± 0.01	97.27 ± 0.02	96.04 ± 0.05		
GDRO	82.70 ± 0.05	80.16 ± 0.45	97.28 ± 0.01	96.10 ± 0.00		
IRM	67.91 ± 0.00	60.20 ± 0.00	96.61 ± 0.00	94.84 ± 0.00		

Figures C6- C15 further show the comparison of various terms of the information decomposition in (1) for each method and dataset. From these results, we observe that the values plotted in the second column (*variation*) and third column (*feature shift*) are similar in magnitude, and based on Equation 1, their difference is close to zero, contributing very little to the overall predictive decomposition. Similarly, we observe that the values plotted in the fourth column (*label shift*) and fifth column (*concept shift*) are similar in magnitude, and based on Equation 1, their difference is close to zero, contributing very little to the overall predictive decomposition. In almost all settings, the difference *conditional informativeness* - *residual* is the majority contributor for the predictive information.

Hyperparameter tuning: We also perform hyperparameter tuning on model parameters. We run a grid search over hyperparameter values and select the best hyperparameter values based on ID validation accuracy (not OOD validation accuracy). We run 1500 experiments (5 datasets, 3 models, and 100 hyperparameter choices sampled from a grid of hyperparameter values). In this case, to get the mean and standard deviation results for accuracies and mutual information terms, we run

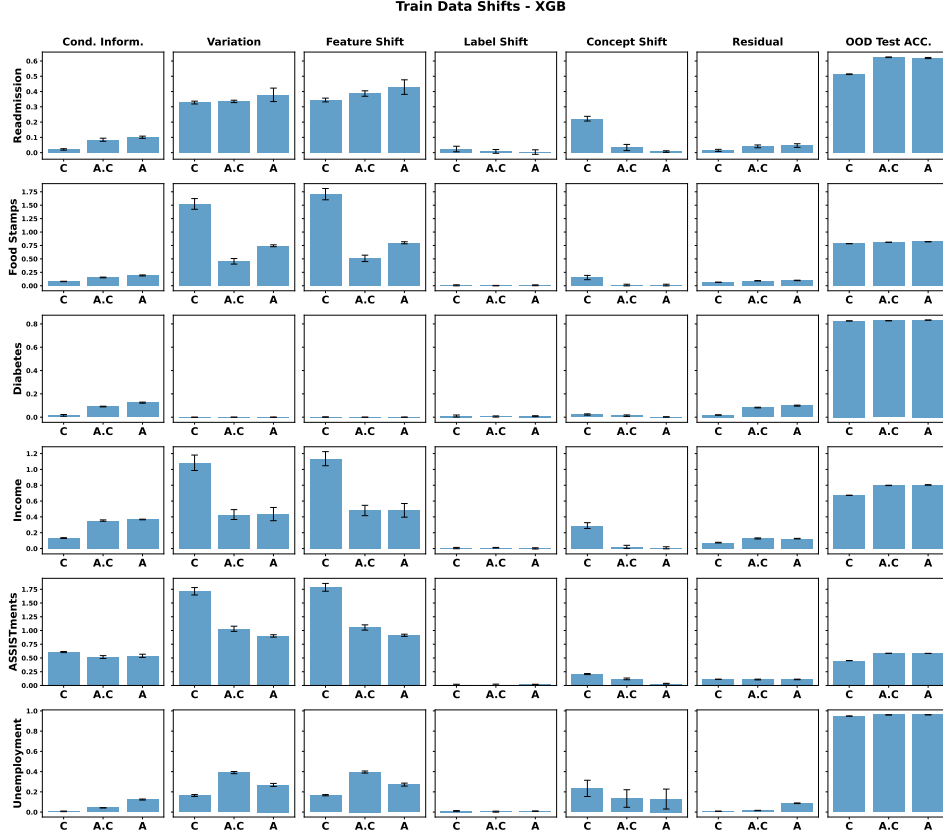


Figure C6: Decomposition of information metrics on train data for XGBoost and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

each experiment for 5 random seeds, even if the effect of random seeds is insignificant in many cases. Results are shown in Figure C4. From Proposition 4.1, the sum of blue and orange terms is positively correlated with the overall model performance. At the same time, due to potential hidden confounding shift in real-world data, the contribution of the difference *conditional informativeness* – *residual* is higher towards the overall predictive information when compared to the contribution of $-\text{variation} / 2 + \text{label shift} / 2 + \text{feature shift} / 2 - \text{concept shift} / 2$. Dataset-specific results shown in Figure C5 and Table C2 also show similar trends.

Potential limitations with predictive information decomposition analysis: While predictive information $I(Y; \hat{Y})$ measures statistical dependence between predictions and labels, it does not guarantee accuracy. For instance, a binary classifier that systematically predicts the *wrong* label (e.g., $\hat{Y} = 1$ when $Y = 0$) achieves maximal $I(Y; \hat{Y}) = 1$ bit (for binary variables) due to perfect anti-correlation, yet yields 0% accuracy. This occurs because mutual information quantifies *reduction in uncertainty* rather than correctness. Other decomposition terms provide further insight: conditional informativeness $I(\phi(X); Y|E)$ may mask harmful variation $I(\phi(X); E|Y)$, while feature shift $I(\phi(X); E)$ can inflate $I(Y; \hat{Y})$ through spurious correlations. Nevertheless, this decomposition remains valuable for analyzing reasonably performant models, as our experiments demonstrate - it reveals whether predictive information stems from generalizable patterns or environment-specific artifacts, enabling targeted improvements while maintaining interpretability.

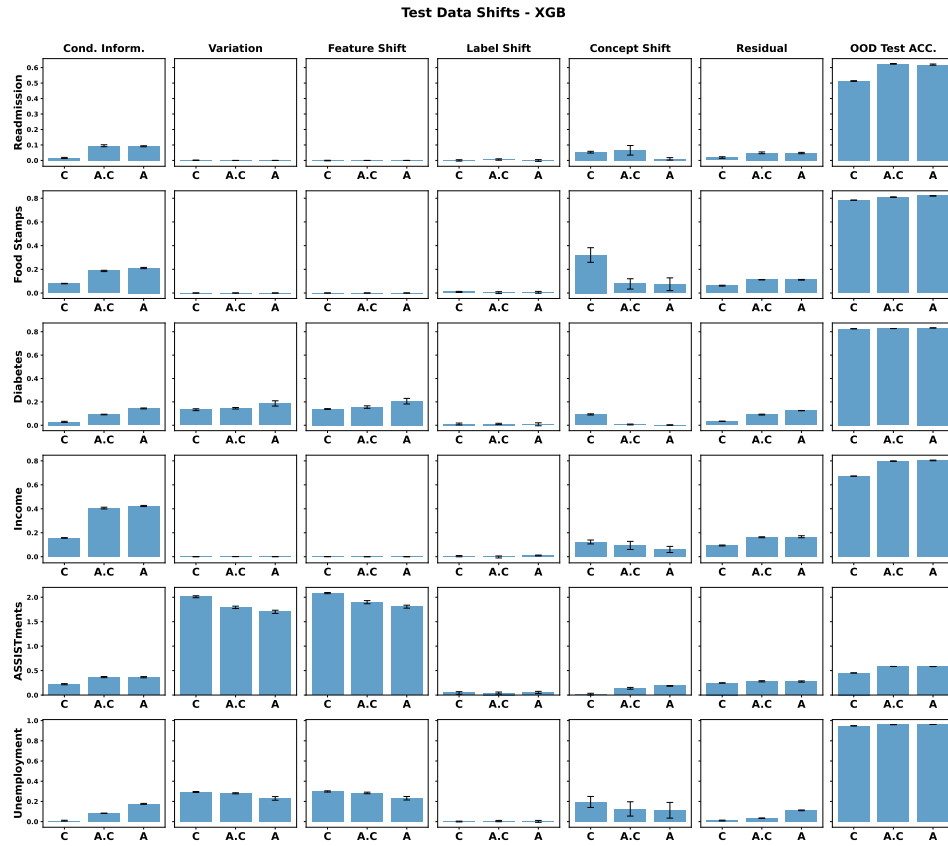


Figure C7: Decomposition of information metrics on test data for XGBoost and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

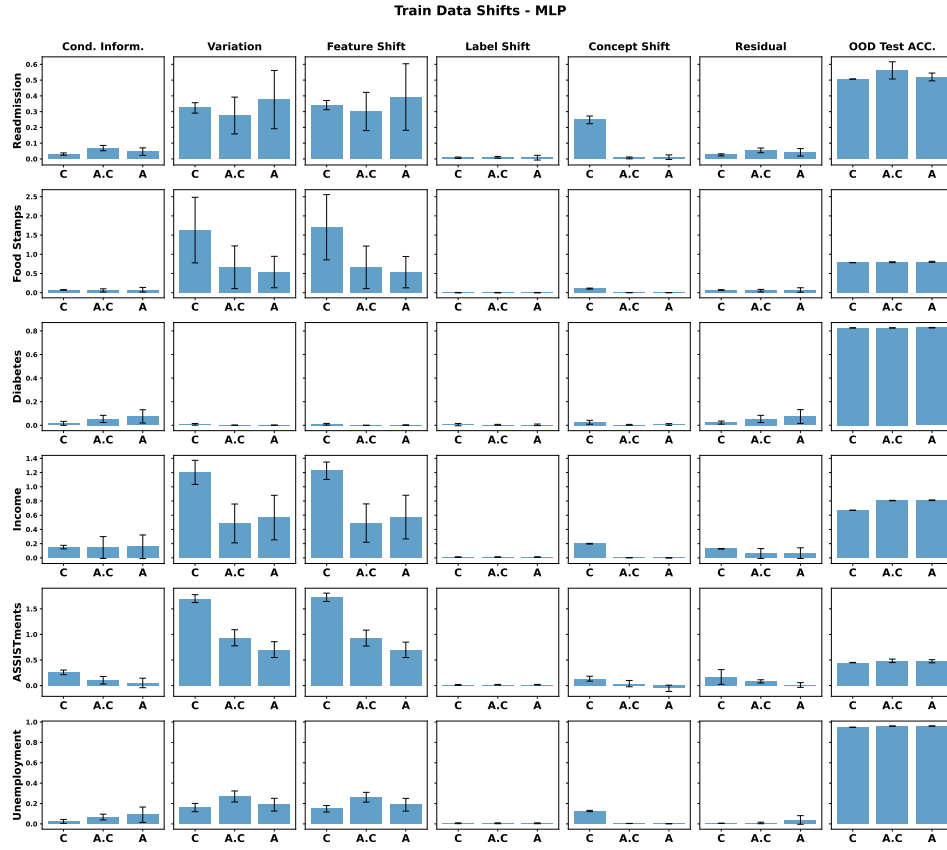


Figure C8: Decomposition of information metrics on train data for MLP and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

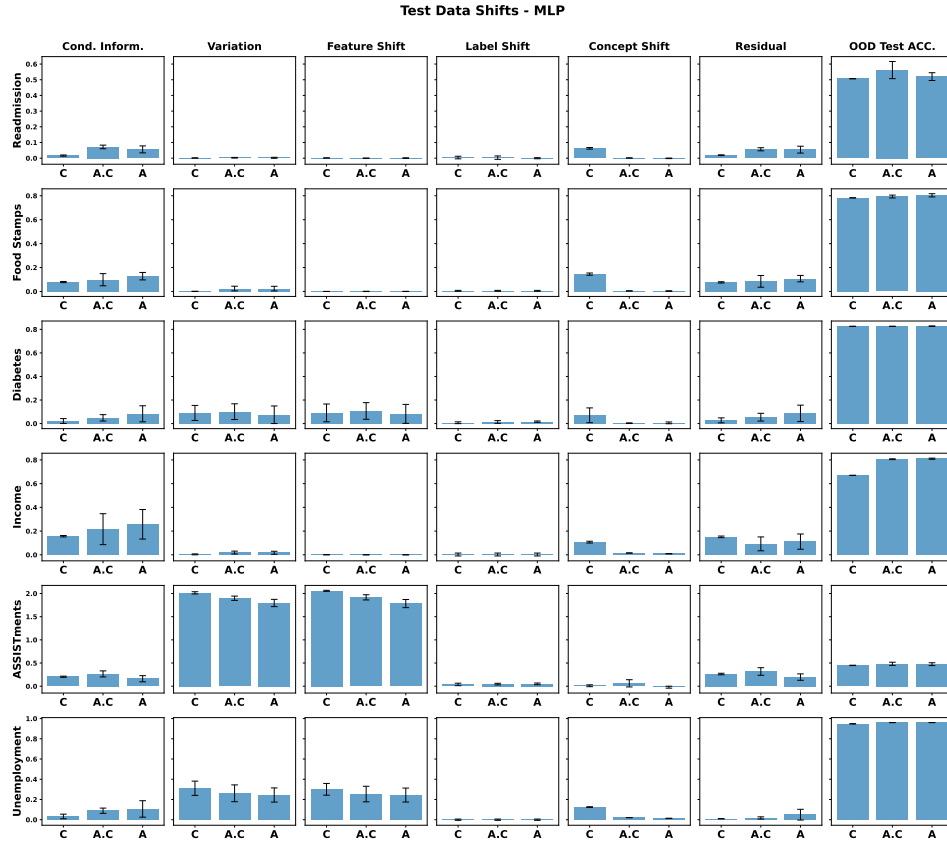


Figure C9: Decomposition of information metrics on test data for MLP and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

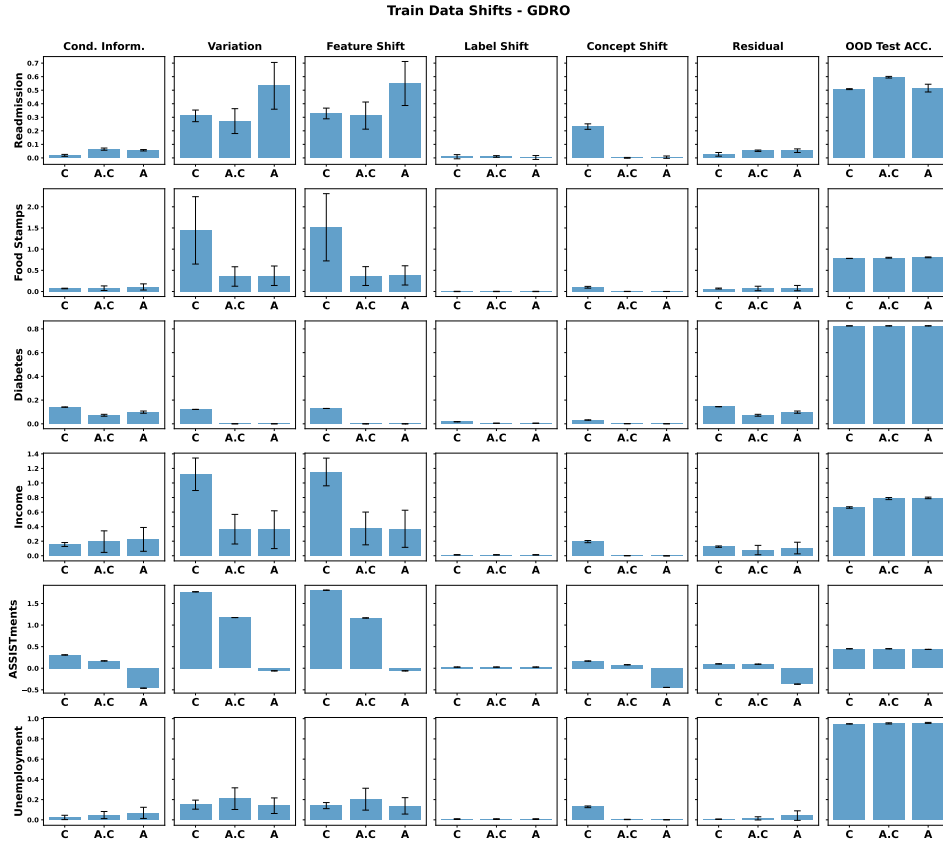


Figure C10: Decomposition of information metrics on train data for GDRO and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

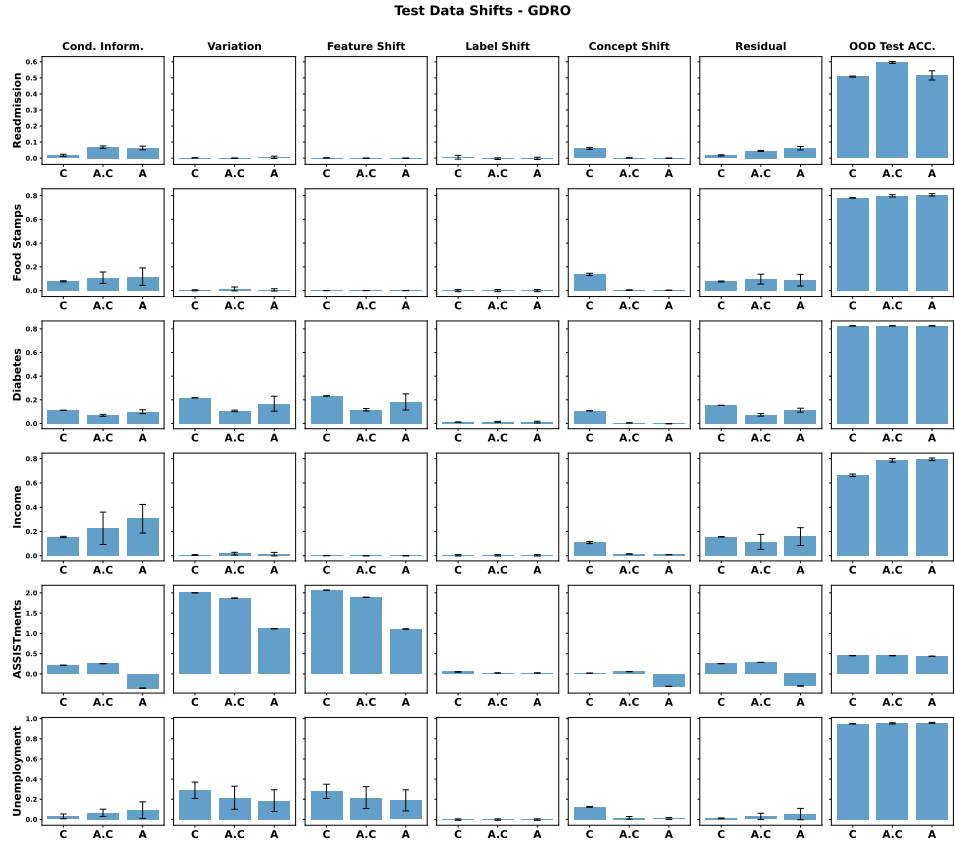


Figure C11: Decomposition of information metrics on test data for GDRO and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

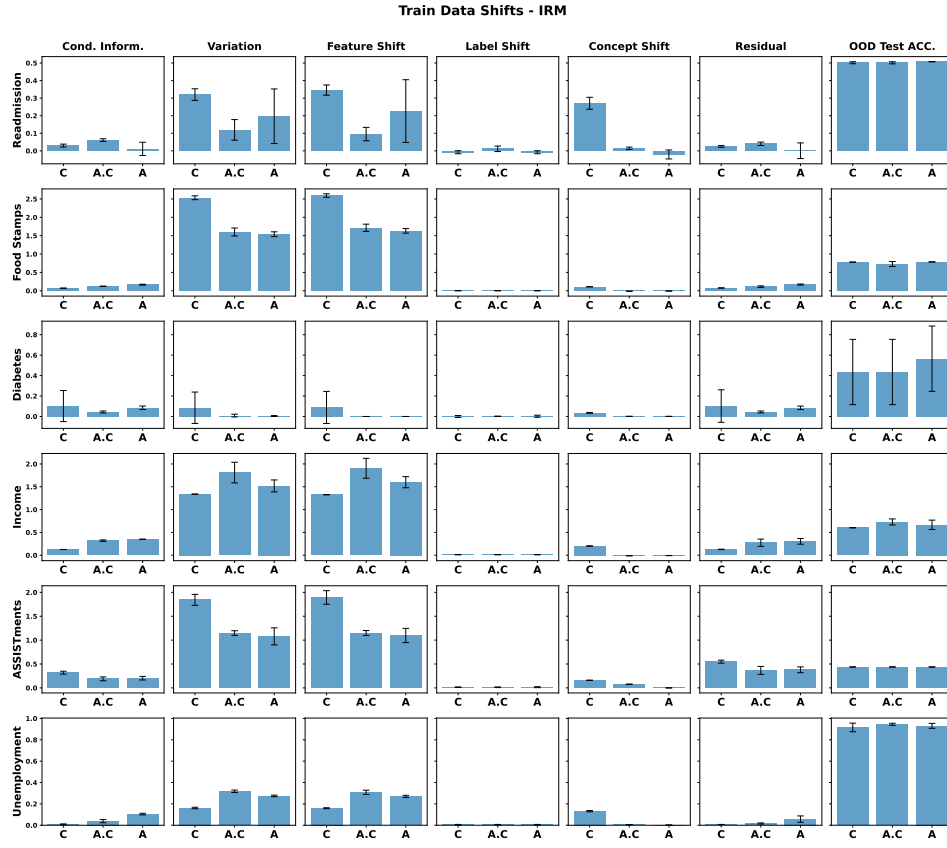


Figure C12: Decomposition of information metrics on train data for IRM and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

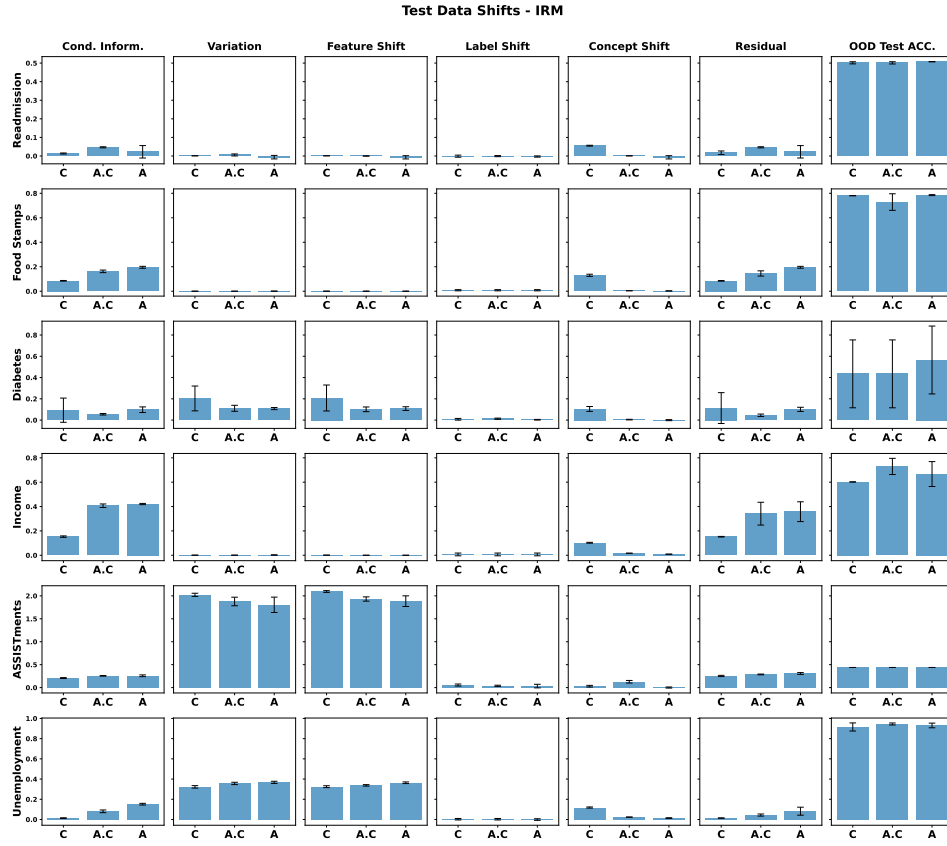


Figure C13: Decomposition of information metrics on test data for IRM and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

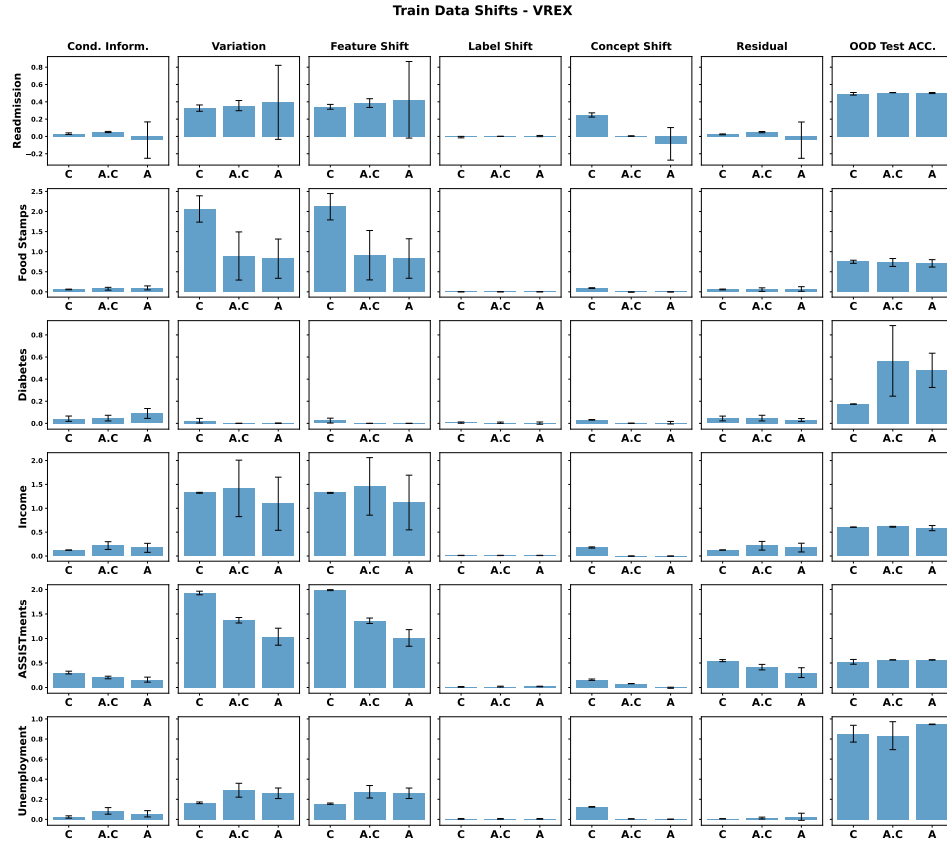


Figure C14: Decomposition of information metrics on train data for VREX and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

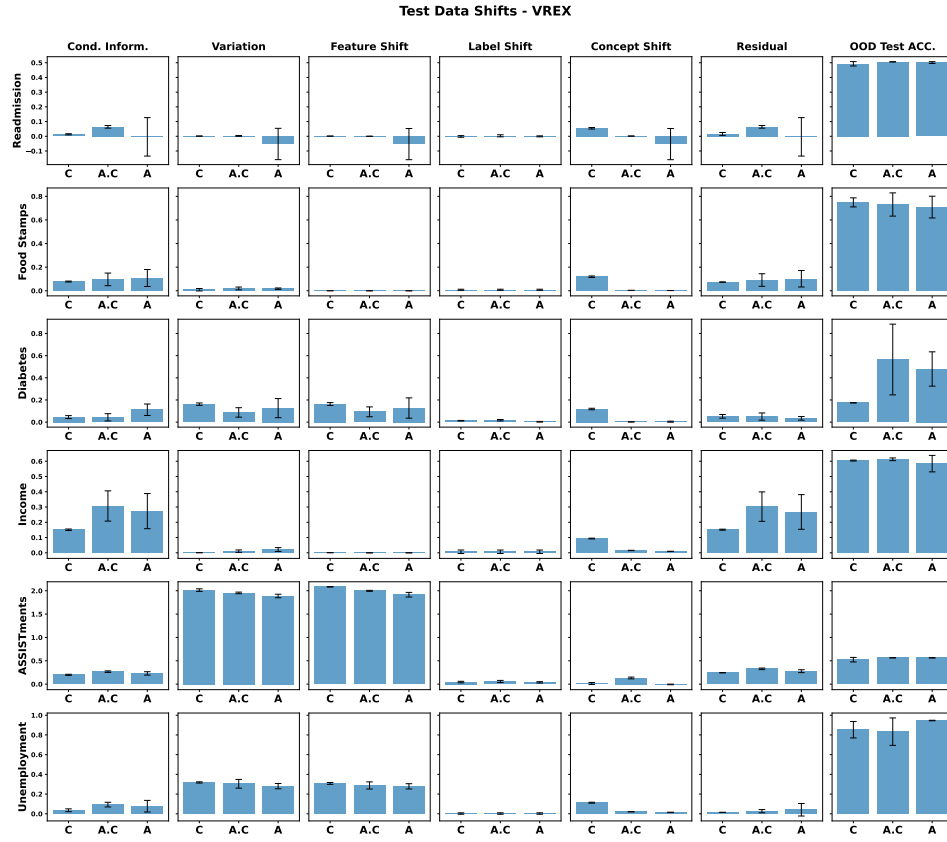


Figure C15: Decomposition of information metrics on test data for VREX and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

814 D Results on synthetic datasets

815 We conduct several experiments on synthetic datasets to investigate: (i) the impact of different
 816 informativeness criteria, (ii) how the level of confounder overlap across environments affects model
 817 performance, and (iii) the influence of the number of causal versus anti-causal variables on the
 818 performance. The data-generating process follows the structural equations in (12), with the causal
 819 structure $U \rightarrow X, U \rightarrow Y, U \rightarrow \mathbf{X}_I$, and $X \rightarrow Y$, where U is an unobserved confounder, X is an
 820 observed covariate, Y is the target variable, and $\mathbf{X}_I$ represents additional informative covariates.

$$\begin{aligned} U &\sim \mathcal{N}(\mu_u^e, \sigma_u) & X &\leftarrow f_X(U) + \mathcal{N}(\mu_x, \sigma_x) \\ X_i &\leftarrow f_i(U) + \mathcal{N}(\mu_i, \sigma_i); X_i \in \mathbf{X}_I & Y &\leftarrow f_Y(X, U) + \mathcal{N}(\mu_y, \sigma_y) \end{aligned} \quad (12)$$

821 We explore different functional forms for f_X , f_i , and f_Y , with the corresponding results presented in
 822 our analysis. Domain shifts are induced by systematically varying the environment-specific mean
 823 parameter μ_u^e of the confounder distribution. This allows us to examine how different degrees of
 824 distributional shift affect model performance across environments.

825 **Informativeness vs. accuracy:** In this set of experiments, we begin with linear structural equations:
 826 $f_X = 0.3U$, $f_i = 0.1U$, and $f_Y = X - 2U$. For these experiments $\mu_u^e \in \{-2, 2\}$ for training
 827 environments and $\mu_u^e \in \{0, 4\}$ for test environments. For the results presented in Figure 5 of the
 828 main paper, we consider one hidden confounding variable with $|\mathbf{X}_I| = 20$ informative covariates.
 829 As we increase the number of informative covariates from 0 to 20, we observe: (i) a reduction in
 830 mean squared error, (ii) improved conditional informativeness, (iii) enhanced feature shift, while
 831 (iv) decreased concept shift. We extend these findings by examining two additional informativeness
 832 criteria. First, we analyze the case with one hidden confounder and a single informative covariate
 833 ($|\mathbf{X}_I| = 1$) across varying noise levels $\sigma_i \in \{0, 0.1, \dots, 2.0\}$. Figure D16 demonstrates that as noise
 834 decreases from 2.0 to 0.0, we observe: (i) reduced mean squared error, (ii) improved conditional
 835 informativeness, (iii) enhanced feature shift, alongside (iv) decreased concept shift. Consistent with
 836 other synthetic experiments, the variation term remains zero throughout. Finally, we investigate a
 837 setting with 20 hidden confounders and corresponding 20 informative covariates, using modified
 838 structural equations $f_X = 0.3U$, $f_i = 0.2U$, and $f_Y = X - 2U$. Figure D17 shows that as we
 839 introduce informative covariates corresponding to hidden confounders, we again observe: (i) reduced
 840 mean squared error, (ii) improved conditional informativeness, (iii) enhanced feature shift, while
 841 (iv) decreased concept shift. These experiments demonstrate how different informativeness criteria
 842 help in achieving OOD generalization.

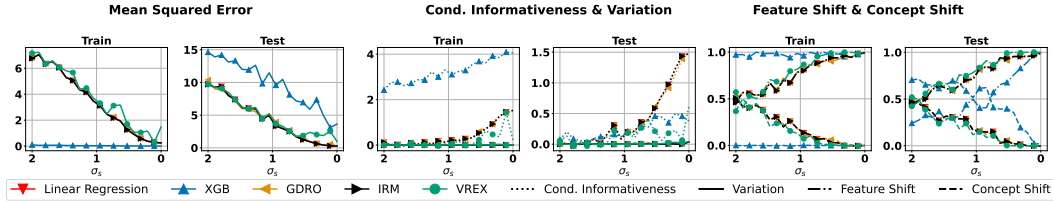


Figure D16: Adding proxy variable with low noise helps in reducing MSE, increasing conditional informativeness and feature shift while reducing concept shift.

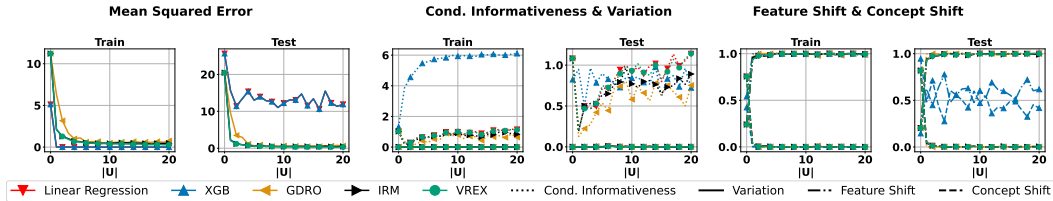


Figure D17: Adding more proxy variables $\mathbf{X}_I$ of a set of hidden confounding variables $\mathbf{U}$ that are informative to Y helps in reducing MSE, increasing conditional informativeness and feature shift while reducing concept shift.

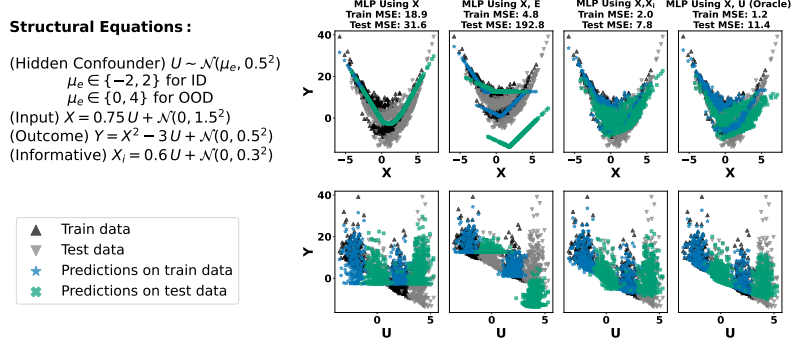


Figure D18: Performance of MLP on the low overlap non-linear setting, for sets of features: (i) X (ii) X and E (environment statistics) (iii) X and X_i (iv) X and U (Oracle).

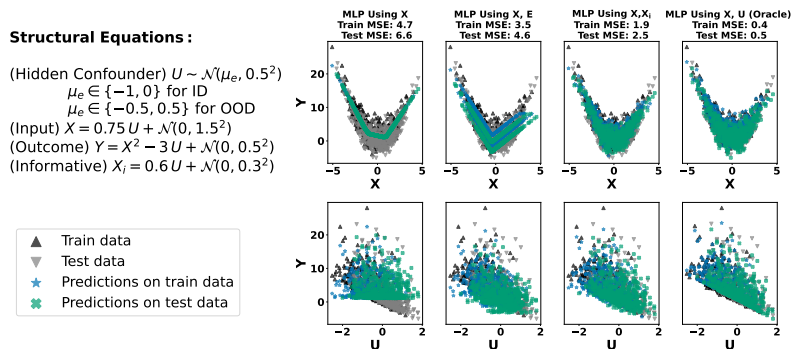


Figure D19: Performance of MLP on the high overlap non-linear setting, for sets of features: (i) X (ii) X and E (environment statistics) (iii) X and X_i (iv) X and U (Oracle).

Overlapping confounder support vs. accuracy: As established in the main paper, recent work by Prashant et al. [62] proposed an OOD generalization method for hidden confounding shift that assumes test confounder support remains within training support. While our results demonstrated successful learning of correct relationships in synthetic linear data when sufficient information about hidden confounders exists in observed covariates, we now extend this analysis to nonlinear data using MLP and XGB models under both low and high confounding support conditions (Figures D18–D21). In low confounding overlap settings (characterized by distant μ_e between ID and OOD data), models relying solely on X learn an inadequate global function that fails to distinguish environments. Performance improves when incorporating environment-specific summary statistics of observed covariates, which helps capture environment-specific relationships, but the most significant gains occur only when leveraging additional informative covariates, underscoring their importance. For completeness, we include both oracle model results and U - Y relationship scatter plots to elucidate the learned input-output mappings. These experiments confirm better performance under high confounding overlap as expected, but more importantly, reveal promising results in the low-overlap regime - a previously unstudied scenario that challenges the common support assumption in the literature.

Number of causal vs anti-causal variables: Recent benchmarks [24] show that baseline methods (e.g., XGB, MLP) consistently match or outperform OOD generalization techniques (e.g., GDRO, IRM, VREX) in real-world datasets, coinciding with the prevalent causal structure $\mathbf{X} \rightarrow Y$ in tabular data. To test if baseline superiority stems from the causal structure: $\mathbf{X} \rightarrow Y$, we vary the causal-to-anti-causal ratio $\rho = |\mathbf{X}_C|/|\mathbf{X}_A| \in \{0.0, \dots, 1.0\}$ with $|\mathbf{X}| = 50$ and $\mathbf{X} = \mathbf{X}_C \cup \mathbf{X}_A$ (e.g., $\rho = 0.5$ gives $|\mathbf{X}_C| = |\mathbf{X}_A| = 25$). Results (Figure D22) show baseline resilience even under the causal structures $\mathbf{X}_C \rightarrow Y \rightarrow \mathbf{X}_A$, though no method dominates universally (Figures D23, D24). Baselines achieve superior PMA-OOD scores [24] (fraction of maximum OOD accuracy). We evaluate: (1) ERM-based methods (XGBoost [15], LightGBM [34], MLP, ResNet [27], SAINT [76], TabTransformer, NODE [61], FT-Transformer, ExpGrad [2]); (2) OOD

Structural Equations :

(Hidden Confounder) $U \sim \mathcal{N}(\mu_e, 0.5^2)$
 $\mu_e \in \{-2, 2\}$ for ID
 $\mu_e \in \{0, 4\}$ for OOD
(Input) $X = 0.75U + \mathcal{N}(0, 1.5^2)$
(Outcome) $Y = X^2 - 3U + \mathcal{N}(0, 0.5^2)$
(Informative) $X_i = 0.6U + \mathcal{N}(0, 0.3^2)$

▲ Train data
▼ Test data
★ Predictions on train data
✱ Predictions on test data

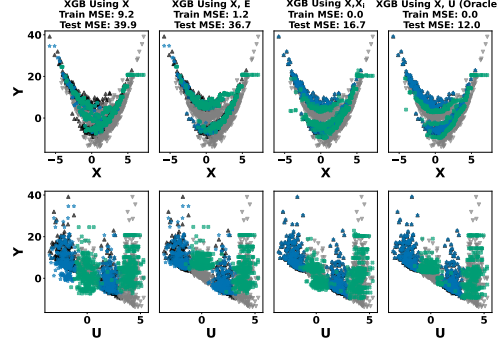


Figure D20: Performance of XGB on the low overlap non-linear setting, for sets of features: (i) X (ii) X and E (environment statistics) (iii) X and X_i (iv) X and U (Oracle).

Structural Equations :

(Hidden Confounder) $U \sim \mathcal{N}(\mu_e, 0.5^2)$
 $\mu_e \in \{-1, 0\}$ for ID
 $\mu_e \in \{-0.5, 0.5\}$ for OOD
(Input) $X = 0.75U + \mathcal{N}(0, 1.5^2)$
(Outcome) $Y = X^2 - 3U + \mathcal{N}(0, 0.5^2)$
(Informative) $X_i = 0.6U + \mathcal{N}(0, 0.3^2)$

▲ Train data
▼ Test data
★ Predictions on train data
✱ Predictions on test data

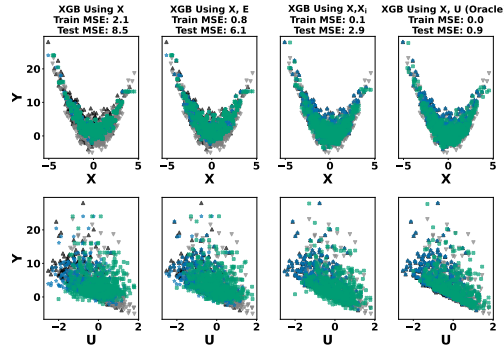


Figure D21: Performance of XGB on the low overlap non-linear setting, for sets of features: (i) X (ii) X and E (environment statistics) (iii) X and X_i (iv) X and U (Oracle).

869 generalization methods (IRM [6], IB-IRM, IB-ERM [30], CausIRL [17], DANN [3], MMD [43],
870 DeepCORAL [81], V-REx [38], Domain Mixup [90, 91]); and (3) Robust optimization methods
871 (DRO [42], GroupDRO [69], Label DRO, Adversarial Label DRO [94]).

Percentage of Max Accuracy (PMA-OOD) on Synthetic Datasets Across All ratios ($\rho = \frac{|X_c|}{|X_t|}$)
Left: Data With Linear Equations, Right: Data With Nonlinear Equations

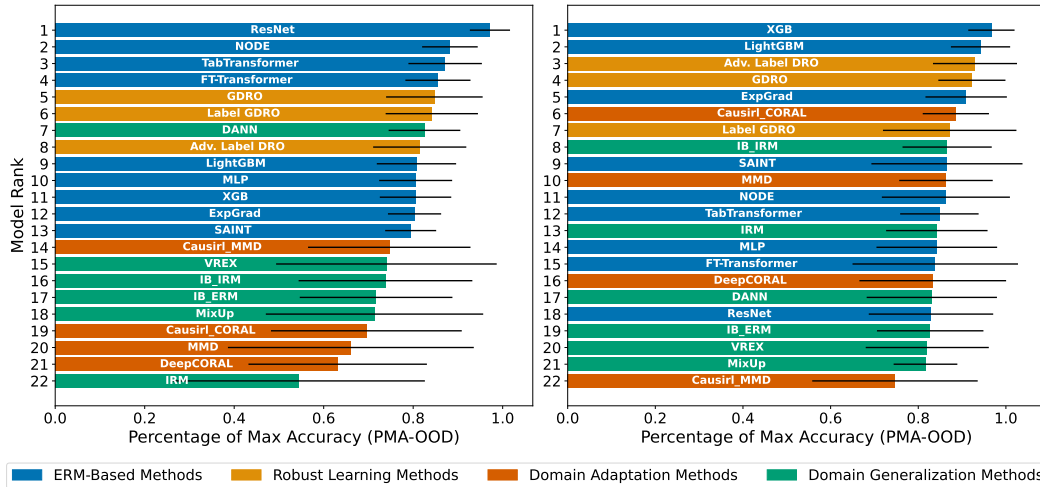


Figure D22: ERM-based methods achieve a better percentage of maximum OOD (PMA-OOD) test accuracy on the synthetic datasets, in line with the real-world results of [24].

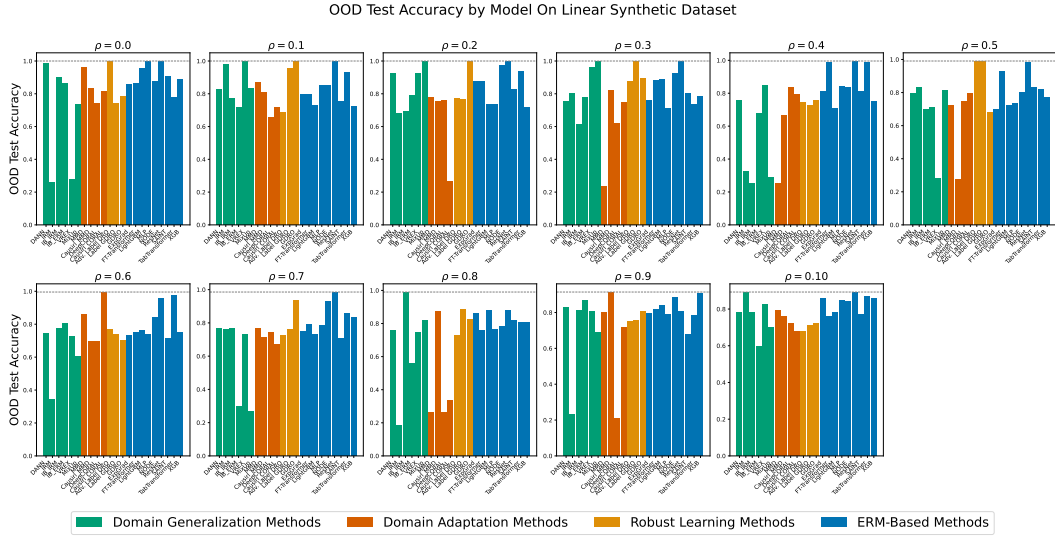


Figure D23: OOD test accuracy on the linear setting per model for different values of ρ .

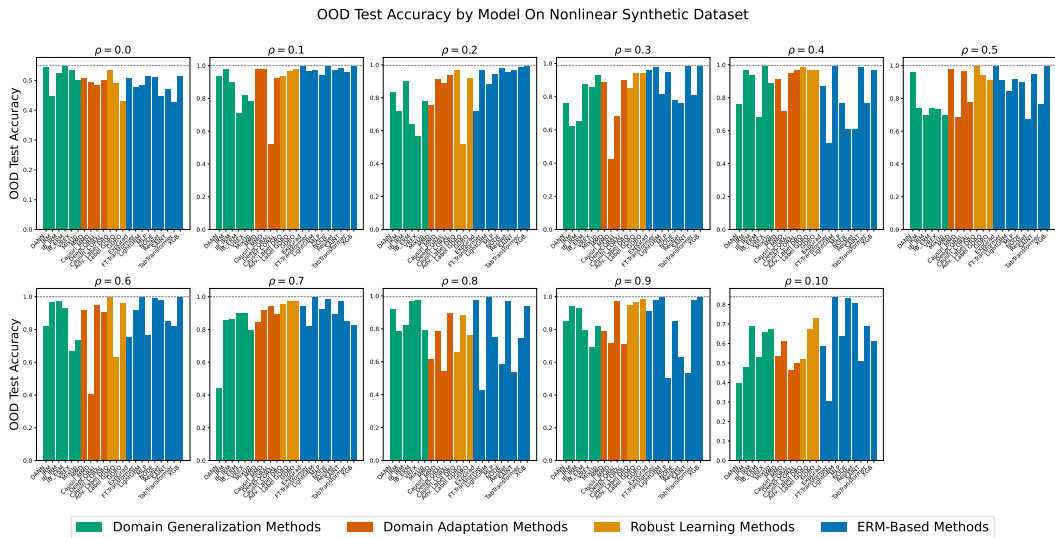


Figure D24: OOD test accuracy on the non-linear setting per model for different values of ρ

872 E Qualitative analysis of hidden confounding shift in real-world datasets

873 In this section, we qualitatively study the presence of hidden confounding shifts in real-world datasets.
874 Tables E3-E18 present potential hidden confounders influencing various observed covariates in
875 several real-world datasets [57]. For additional details regarding these datasets, we refer to [24, 57].
876 We use GPT4o [1] for this task, and we acknowledge that the list of unobserved confounding variables
877 provided is not exhaustive. This study supports the arguments for the existence and impact of hidden
878 confounding shifts across domains in real-world datasets.

879 **Prompt:** We use the prompt below to query GPT4o to get possible hidden confounders.

880 For a target variable, you will be given lists of causal, arguably
881 causal, anti-causal, and spurious covariates. Also, you will be provided
882 with environment variables such that different datasets are collected in
883 environments induced by these environment variables. Provide possible
884 hidden confounders that cause any of the target, causal, arguably causal,
885 spurious, and anti-causal covariates. That is, the hidden confounders
886 should influence the distribution of their children in different domains.
887 Ensure hidden confounders are not a part of the given variables, and your
888 explanation should include how hidden confounders influence their children
889 in different environments.

Table E3: **Dataset:** Food stamps. **Target:** Food stamp reciprocity in the past year for households with child. **Environments:** regions in the United States.

Hidden confounder	Affected variables	Reason for confounding
Local economic conditions	Food stamp reciprocity, Age, Sex, Race, Marital status	Poor economic conditions can increase food insecurity (raising food stamp usage), while also influencing population demographics due to migration, employment patterns, and household formation.
State or regional public assistance policies	Food stamp reciprocity, Marital status, Number of children, Household income	Generous or restrictive assistance policies directly affect food stamp eligibility and indirectly influence household structure and income distribution.
Cultural attitudes towards welfare and public assistance	Food stamp reciprocity, Marital status, Disability, Race, Ethnicity	Social stigma or support for welfare affects both the uptake of food stamps and social norms around marriage, disability reporting, and identity categories.
Regional health care conditions	Food stamp reciprocity, Disability, Cognitive difficulty, Hearing difficulty, Vision difficulty	Regions with limited healthcare access may show higher disability prevalence and greater reliance on food stamps due to increased financial and care burdens.

Table E4: **Dataset:** Income. **Target:** Total person’s income $\geq 56K$ for employed adults. **Environments:** regions in the United States.

Hidden confounder	Affected variables	Reason for confounding
Local economic conditions	Income, Occupation, Educational attainment, Marital status	Stronger local economies offer better-paying jobs and education access, increasing income; they also influence occupational choices and household formation patterns.
State or regional tax and labor policies	Income, Class of worker, Educational attainment, Marital status	Tax incentives and labor protections affect wages and employment types, while also shaping decisions about education and family due to financial security.
Cost of living and regional affordability	Income, Occupation, Marital status, Educational attainment	Higher living costs necessitate higher incomes and may drive occupational or educational shifts; they also influence marriage or cohabitation decisions.
Cultural norms and regional economic history	Income, Educational attainment, Occupation	Cultural values and historical industry presence shape education levels and job opportunities, which in turn affect income distributions.

Table E5: **Dataset:** Public coverage. **Target:** Coverage of non-Medicare eligible low-income individuals. **Environments:** Disability statuses.

Hidden confounder	Affected variables	Reason for confounding
State or regional healthcare policies	Public health coverage, Marital status, Employment status, Income	Differences in Medicaid expansion and public insurance eligibility influence health coverage; these policies also affect economic stability, employment, and family dynamics.
Economic conditions and poverty levels	Public health coverage, Employment status, Income, Marital status	Poorer regions tend to have lower employment rates and incomes, which increase reliance on public health coverage and influence marriage and household composition.
Cultural attitudes towards disability and healthcare	Public health coverage, Disability status, Cognitive difficulty, Hearing difficulty, Vision difficulty	Stigma or support for disability and public care varies culturally, affecting both the reporting of disabilities and the likelihood of seeking or receiving public coverage.
Urban vs. rural divide	Public health coverage, Educational attainment, Marital status, Occupation	Urban areas typically offer better healthcare access, education, and jobs, all of which affect coverage likelihood and socioeconomic indicators.

Table E6: **Dataset:** Unemployment. **Target:** Classify whether a person is unemployed. **Environments:** Educational attainments.

Hidden confounder	Affected variables	Reason for confounding
Local economic conditions	Employment status, Occupation, Marital status, Mobility status	Regional economic strength affects job availability and unemployment rates; it also shapes occupation types, migration decisions, and household stability.
State or regional labor market policies	Employment status, Occupation, Educational attainment, Marital status	Labor regulations and unemployment benefits vary by region, affecting hiring practices, education incentives, and family structures.
Cultural and social norms around employment	Employment status, Disability status, Marital status	Cultural attitudes toward work and dependency influence unemployment reporting and societal roles around disability and family responsibilities.
Health and disability status	Employment status, Disability status, Cognitive difficulty, Hearing difficulty, Vision difficulty	Poor health or disabilities reduce the ability to work, directly increasing unemployment and shaping associated health-related variables.

Table E7: **Dataset:** Voting. **Target:** Classify whether a person voted in the U.S. presidential election. **Environments:** United States census regions.

Hidden confounder	Affected variables	Reason for confounding
Political, family, and peer influences	Voted in national election, Party identification, Political participation	Social networks shape political ideology and engagement, influencing both voting likelihood and party alignment.
Media consumption habits	Voted in national election, Party identification, Political knowledge, Voting behavior	Media exposure affects awareness of political issues and biases, influencing party affiliation, political knowledge, and voting participation.
Social capital	Voted in national election, Political participation, Interest in elections	Strong community ties and civic networks increase political interest and participation, leading to higher voter turnout.
Civic education and political engagement programs	Voted in national election, Interest in elections, Political knowledge	Educational programs raise political awareness and civic responsibility, influencing both knowledge levels and voting decisions.
Historical and cultural context	Voted in national election, Party identification, Interest in elections	Historical events and regional political culture affect interest in elections and party alignment, which in turn influence voting behavior.

Table E8: **Dataset:** Hypertension. **Target:** Whether a person has hypertension. **Environments:** Body Mass Index (BMI) values.

Hidden confounder	Affected variables	Reason for confounding
Socioeconomic status	Income, Employment status, Smoking habits, Alcohol consumption, Physical activity, Healthcare access, Medical cost, High blood pressure diagnosis	Lower socioeconomic status reduces access to healthcare and healthy lifestyle options, leading to poor diet, limited activity, and delayed diagnosis, which jointly influence both BMI and hypertension risk.
Access to healthcare services	Healthcare access, Medical costs, Smoking habits, Alcohol consumption, Physical activity, High blood pressure diagnosis	Limited healthcare access results in underdiagnosis and unmanaged hypertension, while also affecting lifestyle choices that vary with BMI, confounding the relationship between BMI and hypertension.
Genetic predisposition	Age group, Race, Sex, Smoking habits, Diabetes, High blood pressure diagnosis	Genetic risk factors for hypertension may co-vary with demographic attributes and influence both hypertension prevalence and BMI distribution across subpopulations.
Psychosocial stress	Smoking habits, Alcohol consumption, Physical activity, High blood pressure diagnosis, Diabetes, Age group	Chronic stress alters behavior (e.g., smoking, inactivity) and physiological responses, contributing to both increased BMI and elevated blood pressure, thus confounding the BMI–hypertension link.
Environmental factors	Physical activity, Diet, Smoking habits, High blood pressure diagnosis, Diabetes, BMI category	Living environments affect access to recreational spaces, food quality, and pollution exposure, influencing both BMI and hypertension risks. These vary across BMI categories, creating confounding.
Dietary habits beyond fruits and vegetables	Alcohol consumption, Smoking habits, Physical activity, High blood pressure diagnosis, BMI category	High-sodium or processed-food diets raise both BMI and hypertension risk. Variation in such unmeasured dietary habits across BMI categories creates spurious associations with hypertension.

Table E9: **Dataset:** College Scorecard. **Target:** Predict completion rate for first-time, full-time students at four-year institutions. **Environments:** Based on Carnegie Classifications.

Hidden confounder	Affected variables	Reason for confounding
Institutional funding and re-sources	Accreditor, Control of institution, Highest degree awarded, In-state tuition, Out-of-state tuition, Cost of attendance, SAT scores	Wealthier institutions can offer better academic support, facilities, and programs, leading to higher completion rates and more selective admission profiles. This varies across Carnegie classifications.
Regional socio-economic factors	Region, Poverty rate, Unemployment rate, SAT scores	Economic conditions across regions affect affordability, student preparedness, and institutional support levels, all influencing both enrollment outcomes and graduation likelihood.
Demographic factors	HBCU flag, Federal loan recipient rate, Pell grant recipient rate, ACT scores, Undergraduate enrollment	Student demographics shape financial aid needs, academic preparation, and graduation rates. Their effect differs by institution type and selectivity under Carnegie categories.
Community support and engagement	Distance-education flag, Federal loan recipient rate, Pell grant recipient rate, SAT scores, Undergraduate enrollment	Supportive institutional communities improve retention and completion. Variation in engagement across institution types and student aid profiles induces confounding.
Admission selectivity	Admission rate, SAT (reading/math) midpoints, ACT midpoint, Undergraduate enrollment	Selective admissions correlate with better-prepared students and higher completion rates, and vary with institutional prestige and classification.
State and local policies	Region, Poverty rate, Unemployment rate, Cost of attendance	Differences in education funding and public policy affect cost structures and completion outcomes, interacting with institutional classification and regional demographics.

Table E10: **Dataset:** ASSISTments. **Target:** Predict whether a student solves a problem correctly on the first attempt in an online learning tool. **Environments:** Different schools.

Hidden confounder	Affected variables	Reason for confounding
Institutional teaching quality	Hint count, Attempt count, Skill ID, Problem type, Tutor mode, Position, Type, First action, Milliseconds to first response, Overlap time, Average confidence	Variation in instructional quality and pedagogy across schools affects how effectively students engage with content, leading to differences in problem-solving strategies, response behavior, and emotional states.
Student motivation	Hint count, Attempt count, Skill ID, Problem type, First action, Milliseconds to first response, Overlap time, Average confidence	Differences in intrinsic motivation across schools influence students' willingness to persevere, seek help, or give up quickly, affecting interaction and performance.
Classroom environment	Hint count, Attempt count, Tutor mode, Position, Type, First action, Average confidence	Peer dynamics, classroom culture, and noise levels vary by school and affect how confidently and independently students solve problems.
School technology infrastructure	Hint count, Attempt count, Tutor mode, Position, Type, First action, Milliseconds to first response, Overlap time, Average confidence	Access to reliable devices and fast internet differs by school, influencing response time, tool usage, and student experience.
Teacher-student interaction	Hint count, Attempt count, Tutor mode, Position, Type, First action, Milliseconds to first response, Overlap time, Average confidence	The level of teacher guidance and feedback shapes how much support students require during problem-solving, affecting engagement and confidence.
Previous academic performance	Hint count, Attempt count, Skill ID, Problem type, First action, Average confidence	Students' prior achievement affects how easily they solve problems, their need for assistance, and their confidence, all of which vary across schools.

Table E11: **Dataset:** ICU. **Target:** Predict whether the patient will stay in the ICU for longer than 3 days. **Environments:** Insurance types.

Hidden confounder	Affected variables	Reason for confounding
Socioeconomic status (SES)	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, etc.	SES influences access to healthcare, preventive services, and overall health status. Differences in SES across insurance types lead to variability in pre-ICU health, physiological indicators, and ICU stay duration.
Hospital resources and care quality	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, etc.	Hospital infrastructure and care standards affect monitoring, intervention speed, and clinical decisions. These factors vary by insurance coverage and influence ICU outcomes and vitals.
Comorbidities	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, etc.	Presence of chronic conditions (e.g., diabetes, cardiovascular disease) affects both the need for prolonged ICU care and physiological measurements. The distribution of comorbidities differs across insurance types.
Insurance-related treatment variability	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, etc.	Differences in treatment timing, intensity, and access to specialists based on insurance policies affect ICU stay duration and clinical metrics.
Genetic factors	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, etc.	Inherited traits influence predisposition to organ failure, metabolic responses, and recovery trajectories. These effects are partially mediated by ethnicity and age distributions, which vary across insurance groups.

Table E12: **Dataset:** Hospital mortality. **Target:** Classify whether an ICU patient expires in the hospital during their current visit. **Environments:** Insurance types.

Hidden confounder	Affected variables	Reason for confounding
Socioeconomic status (SES)	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Systolic blood pressure, White blood cell count, etc.	SES shapes access to timely and high-quality care, preventive services, and general health status. Patients with higher SES often have better insurance and outcomes, leading to confounding with mortality risk.
Comorbidities	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, Systolic blood pressure, etc.	Pre-existing conditions such as diabetes or heart disease increase mortality risk and influence physiological features. Their prevalence differs by insurance type, creating confounding.
Hospital resources and care quality	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, Systolic blood pressure, White blood cell count, etc.	Access to advanced treatments, trained staff, and timely interventions influences survival rates. These factors correlate with insurance coverage, confounding mortality outcomes.
Genetic factors	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, etc.	Genetic predispositions affect disease susceptibility and treatment responses. Variations in genetic risk factors may correlate with demographic traits across insurance types.
Lifestyle and behavioral factors	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, Temperature, Systolic blood pressure, White blood cell count, etc.	Behaviors such as smoking, diet, and physical activity affect long-term health and mortality risk. These behaviors vary systematically with SES and insurance coverage, influencing both target and physiological features.

Table E13: **Dataset:** Childhood lead. **Target:** Predict blood lead levels above CDC blood level reference value. **Environments:** Poverty-income ratios.

Hidden confounder	Affected variables	Reason for confounding
Environmental Exposure	Blood lead levels, Country of birth, Age, Race and Hispanic origin	Environmental exposure to lead influences blood lead levels, and this varies significantly across socio-economic groups. People in lower PIR groups are more likely to live in areas with higher lead contamination, which contributes to higher blood lead levels. Furthermore, environmental factors may affect the demographic distribution (e.g., country of birth, race).
Access to Healthcare	Blood lead levels, Age, Gender, Race and Hispanic origin, Marital status, Education	Limited access to healthcare, especially in lower PIR groups, means fewer opportunities for detection and treatment of lead poisoning. This results in higher blood lead levels, with disparities also influencing demographic variables like age, gender, and education. Additionally, healthcare access varies by insurance and socio-economic status, further confounding the relationships.
Diet and Nutrition	Blood lead levels, Age, Gender, Race and Hispanic origin	Dietary factors, such as poor nutrition in lower PIR groups, can exacerbate lead absorption. Malnutrition increases the body's susceptibility to lead poisoning, raising blood lead levels. In contrast, higher PIR groups may have better access to nutritious foods, lowering lead absorption, thus creating a confounding effect in how socio-economic status and race influence lead toxicity.
Housing Conditions	Blood lead levels, Country of birth, Race and Hispanic origin, Marital status	Older housing conditions, which are more prevalent in lower PIR groups, contribute significantly to elevated lead exposure (e.g., lead paint, poor plumbing). These living conditions directly influence blood lead levels and can also correlate with demographic factors like country of birth, race, and marital status. This introduces confounding, as socio-economic status impacts both exposure and the demographics of affected individuals.
Occupation	Blood lead levels, Age, Race and Hispanic origin, Marital status, Education	Certain occupations, which are more common among lower PIR groups, involve higher lead exposure (e.g., construction, manufacturing). Occupational lead exposure directly impacts blood lead levels and is often correlated with education, marital status, and socio-economic status. The varying prevalence of lead exposure by occupation introduces confounding, especially across different PIR groups.

Table E14: **Dataset:** Diabetes. **Target:** Predict diabetes. **Environments:** Preferred race categories.

Hidden confounder	Affected variables	Reason for confounding
Genetic predisposition	Diabetes, BMI, High blood pressure, High blood cholesterol	Genetic factors and family history contribute to both the onset of diabetes and comorbid conditions like obesity, hypertension, and high cholesterol. These genetic predispositions can make individuals more susceptible to diabetes, leading to confounding as they correlate with other health indicators.
Access to healthcare	Diabetes, Physical health, BMI, Healthcare coverage, Health checkups	Limited or unequal access to healthcare, especially in marginalized racial groups, leads to disparities in diabetes diagnosis, management, and comorbidity treatment. It also influences the frequency of health checkups and access to medications, which can confound the relationship between diabetes status and other health metrics.
Dietary habits and food availability	Diabetes, BMI, Physical health, Alcohol consumption, Fruit and vegetable intake	Dietary habits, often shaped by socioeconomic status and local food environments, influence weight, health behaviors (such as alcohol consumption), and diabetes risk. People in lower socioeconomic strata may have limited access to healthy food options, leading to higher BMI and increased diabetes risk, creating confounding effects on health outcomes.
Psychosocial stress and mental health factors	Diabetes, Mental health, Physical health, BMI, Physical activity, Doctor visits	Chronic stress and mental health issues, often higher in marginalized groups, contribute to diabetes development and complicate its management. These factors also affect physical health (e.g., weight gain due to stress) and health-seeking behaviors (e.g., fewer doctor visits), leading to confounding by influencing both diabetes risk and its associated variables.
Socioeconomic status beyond income	Diabetes, Income, Physical health, Healthcare coverage, Education level	Socioeconomic factors, such as occupation, education, and neighborhood wealth, influence access to healthcare, nutrition, and overall health behaviors. These factors can confound the relationships between diabetes and other socio-economic variables like income and education, as they shape opportunities for prevention and treatment.

Table E15: **Dataset:** Sepsis. **Target:** Predict, from a set of fine-grained ICU data, whether a patient will experience sepsis onset within the next 6 hours. **Environments:** Lengths of ICU stay.

Hidden confounder	Affected variables	Reason for confounding
Infection prevalence in ICU	SepsisLabel, Temperature (Temp), Leukocyte count (WBC), Heart rate (HR), Blood urea nitrogen (BUN)	Higher infection rates in certain ICU units can lead to a higher probability of sepsis onset (SepsisLabel). These infection rates influence biomarkers such as WBC, HR, and BUN, creating confounding because the unit's infection environment affects both the likelihood of sepsis and the observed clinical measures.
Quality of ICU Care	SepsisLabel, Fibrinogen concentration (Fibrinogen), Leukocyte count (WBC), Platelet count (Platelets)	Higher-quality care in certain ICUs may lead to earlier identification and treatment of sepsis, resulting in more accurate SepsisLabel predictions. Additionally, better care could affect biomarkers like fibrinogen, WBC, and platelets, which are critical in sepsis detection and progression, thereby confounding the relationships between these variables and the outcome.
Severity of underlying conditions	SepsisLabel, Blood urea nitrogen (BUN), Creatinine, Lactate, Calcium	Patients with severe chronic conditions (e.g., kidney disease, cardiovascular issues) are at a higher risk of sepsis and may show abnormal levels in biomarkers like BUN, creatinine, lactate, and calcium. These underlying conditions contribute to the SepsisLabel outcome and confound the relationship between the biomarkers and the likelihood of sepsis, varying across ICU units depending on patient population.
Patient's socio-economic status	Age (Age), Gender (Gender), Leukocyte count (WBC), Fibrinogen concentration (Fibrinogen), Platelet count (Platelets)	Socio-economic factors, such as access to healthcare, can influence both the likelihood of sepsis and the observed clinical biomarkers. For example, patients from lower socio-economic backgrounds may have delayed hospitalizations or inadequate care, which affects both SepsisLabel and the progression of sepsis as indicated by WBC, fibrinogen, and platelet levels.
Hospital-specific protocols and treatment guidelines	SepsisLabel, Lactate, Glucose, Creatinine	Differences in hospital protocols for sepsis treatment, such as timing of interventions and choice of sepsis bundles, can affect both the SepsisLabel and biomarkers like lactate, glucose, and creatinine. These protocols lead to variability in how sepsis is diagnosed and treated across different hospitals and ICU units, confounding the relationship between biomarkers and sepsis outcomes.

Table E16: **Dataset:** Hospital readmission. **Target:** Predict whether a diabetic patient is readmitted to the hospital within 30 days of their initial release. **Environments:** Admission sources.

Hidden confounder	Affected variables	Reason for confounding
Socio-economic status (SES)	Race, Gender, Age, Payer code, Medical specialty, Number of outpatient visits, Number of emergency visits, Number of inpatient visits, Diabetes medication prescribed	SES influences access to healthcare, patient demographics, and chronic disease rates. It can also determine the type of care received based on admission source (e.g., emergency department vs. outpatient settings). Differences in healthcare access, such as availability of medications, may impact readmission rates and associated variables like outpatient visits and prescribed medication.
Severity of illness	Primary diagnosis, Secondary diagnosis, Number of diagnoses, Discharge type, Medication changes (e.g., Insulin, Glipizide)	More severe illness increases the likelihood of readmission and influences the complexity of diagnoses and the treatments administered. The severity of illness may differ based on the admission source (e.g., emergency versus outpatient), impacting the number and type of diagnoses and treatments prescribed at discharge, affecting readmission likelihood.
Access to healthcare resources	Time in hospital, Discharge disposition, Number of procedures, Number of medications, Number of lab tests	Access to healthcare resources (e.g., time in hospital, availability of procedures and medications) influences treatment decisions and outcomes. Different admission sources may have varying levels of available resources, leading to different lengths of stay, the types of procedures performed, and overall treatment quality, which can affect the likelihood of readmission.
Patient's adherence to medication	Change in medications, Diabetes medication prescribed, Number of outpatient visits, Number of emergency visits	Adherence to prescribed medications is often influenced by socio-economic status, which can vary depending on admission source. Non-adherence may lead to medication changes, affecting diabetes control and subsequent readmission risk. The number of outpatient and emergency visits can also reflect how well a patient manages their diabetes and the likelihood of complications.
Hospital-specific protocols	Time in hospital, Number of diagnoses, Discharge disposition, Readmitted	Different hospitals and healthcare systems implement various protocols for discharge planning and readmission prevention, which can affect readmission rates. These protocols may vary by admission source, where patients admitted via the emergency department may receive different follow-up instructions and care than those admitted through other channels.

Table E17: **Dataset:** MEPS. **Target:** Measure of health care utilization. **Environments:** Insurance types.

Hidden confounder	Affected variables	Reason for confounding
Socioeconomic status (SES)	Years of education, Employment status, Hourly wage, Paid sick leave, Paid leave to visit doctor, Family size, Insurance coverage, Healthcare utilization	SES can influence access to healthcare services, insurance coverage, and the ability to use medical services. Insurance types often correlate with SES levels, where individuals with lower SES may be more likely to be insured by government programs (e.g., Medicaid), which in turn impacts healthcare utilization patterns across different SES groups.
Healthcare access	Paid sick leave, Insurance coverage, Employer offers health insurance, Healthcare utilization	Limited access to healthcare, such as lack of insurance or paid sick leave, directly impacts healthcare utilization. The type of insurance a person holds is often tied to access to various medical services. The level of coverage and accessibility differs across insurance types, influencing healthcare behaviors such as whether a patient can afford and utilize healthcare services.
Health behaviors	Perceived health status, Asthma medications, Limitations in physical functioning, Healthcare utilization	Lifestyle factors like smoking, alcohol consumption, and physical activity can directly affect health status and healthcare needs. Health behaviors differ across groups with different insurance types, and these behaviors contribute to healthcare utilization. Insurance coverage can also be influenced by perceived health status, which varies across insured groups, affecting utilization of medical services.
Chronic health conditions	Asthma medications, Perceived health status, Limitations in physical functioning, Healthcare utilization	Chronic conditions like asthma or diabetes increase the need for healthcare services, leading to higher healthcare utilization. People with chronic conditions are more likely to be covered by Medicare or Medicaid, which influences the type of insurance they hold and impacts their healthcare utilization and associated features (e.g., medications and physical limitations).
Regional healthcare infrastructure	Region, Family size, Healthcare utilization, Paid sick leave, Paid leave to visit doctor, Employment status	The quality and availability of healthcare infrastructure vary across regions, which can impact healthcare utilization. In underserved regions, people may be more reliant on public insurance options like Medicaid, affecting healthcare access and behaviors. The regional differences in healthcare systems can lead to disparities in access to medical services, insurance coverage, and utilization of healthcare.

Table E18: **Dataset:** Poverty. **Target:** Predict household income-to-poverty ratio. **Environments:** Citizenship statuses.

Hidden confounder	Affected variables	Reason for confounding
Social capital	Income, Unemployment compensation, Disability benefits, Family size, Housing conditions, Household income-to-poverty ratio	Stronger social support networks can improve access to income, benefits, and resources like unemployment or disability compensation. Social capital can vary based on citizenship status, which in turn influences income, housing conditions, and eligibility for benefits, confounding the relationship between household income and poverty ratios.
Workplace discrimination or bias	Household income, Worker's compensation, Unemployment compensation, Pension, Family income, Disability benefits	Discrimination in the workplace can limit opportunities for higher wages, pensions, or disability benefits, especially for marginalized groups (e.g., racial minorities, non-citizens). This bias varies by citizenship status and leads to biased associations between income, benefits, and poverty levels.
Healthcare access	Household income, Health insurance premiums, Medical expenses, Disability benefits, Family size, Health conditions, Medicaid/Medicare assistance	Limited access to healthcare, particularly for non-citizens, can lead to higher medical expenses and greater reliance on public health assistance. Citizenship status directly impacts eligibility for public healthcare programs, confounding relationships between medical expenses and income, and affecting poverty ratios.
Access to education and skills development	Educational attainment, Household income, Employment status, Unemployment compensation, Income from assistance, Savings	Disparities in educational opportunities, often linked to citizenship status, influence income potential and eligibility for government assistance. These disparities, in turn, affect the household income-to-poverty ratio and employment outcomes, creating confounding relationships between education and income.
Cultural factors	Family size, Household income, Savings, Disability benefits, Healthcare utilization, Income assistance, Living arrangements	Cultural factors influence financial management, family support, and the use of social programs. These factors can vary across citizenship statuses, leading to differences in how household income and assistance are distributed, and thus confounding the relationship between income, benefits, and financial behavior.
Housing market and rent conditions	Housing ownership, Housing costs, Family size, Household income, Rent payments, Social security benefits, Medical aid	Housing market conditions, especially rent disparities, can create financial strain and impact the household income-to-poverty ratio. Citizenship status often influences eligibility for housing assistance and rent subsidies, which confounds the relationship between housing costs and income, particularly in areas with significant immigrant populations.